

Adaptive Design of Clustered Experiments

Carlos Gonzalez-Perez*

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Abstract

I propose two algorithms that implement near-optimal designs in clustered surveys and experiments without previous knowledge of the intra-cluster correlation (ICC). Moreover, under a symmetry condition, I show that inference on sample averages is invariant to adaptive data collection protocols like mine which only depend on sample variances. A researcher must decide how many clusters and how many units per cluster to sample in order to minimize the variance of a population average estimator under a fixed budget and linear costs. Sampling additional clusters reduces the statistical dependence between observations, but clusters are often costly to recruit or administer, what creates a trade-off between statistical accuracy and costs mediated by the ICC. I show that a two-stage explore-then-commit algorithm and an improved adaptive algorithm both achieve negligible excess variance compared to the oracle design with known ICC. I further show that adaptive data collection depending on sample variances (as the proposed algorithms) do not compromise inference: the resulting estimators are asymptotically independent of the design, hence, the analyst can keep all the data from the pilot and preserve asymptotic normality without any adjustments. Finally, I show that my algorithms are minmax optimal up to logarithmic factors and strictly improve upon any misspecified static policy. These results provide a foundation for adaptive sampling in clustered surveys and randomized controlled trials.

1 Introduction

Over the last few years, surveys and Randomized Control Trials (RCT) have become increasingly larger, not only in size but in scope, spanning several regions in the quest for external validity and efficiency (Muralidharan and Niehaus (2017)).¹ Multi-cluster designs are prevalent in the Survey and Applied Economics literature, including applications to development economics, education, health, finance, trade, gender and more (Duflo and Banerjee (2017)). For instance, Mbiti et al. (2019) deploy an experiment at the school level consisting of a combination of school grants and teacher incentives. The implementation featured random selection of 10 districts from mainland Tanzania and then, random treatment allocation to 35 schools in each district. Analogously, consider the survey in Sedgwick (2015), in which in-person interviews were conducted to learn about the preferences of the British household about the use of medical data. The designer randomly selected a few postcodes in the UK

*University of Oxford, Department of Economics. Email: carlos.gonzalezperez@economics.ox.ac.uk. I would like to thank Maximilian Kasy, Davide Viviano, Rahul Singh and participants at the Econometrics Seminar at Harvard for feedback on this manuscript.

¹Regions can be physical or figurative. Examples of physical regions include countries, provinces, villages, etc. and examples of figurative regions include demographic strata, markets, industries, etc.

(proportionally to their size) and then interviewed a few households from each postcode.² We refer to these experimental designs as clustered experiments, where the design is given by the width (number of clusters, K) and depth (number of units per cluster, N).

While the sample average of the estimators obtained via clustered experiments remain unbiased conditional on sequential randomization, their variance depend critically on the design. From a statistical perspective, the inclusion of multiple clusters is always desirable. If the researcher believes that observations/treatment effects are more correlated within clusters than across clusters, she can always improve on the variance of the resulting estimator by sampling from several clusters. However, the inclusion of additional clusters is often costly, especially in the context of developing countries. Some examples of cluster-level fixed costs are travel costs, training of additional staff, time-costs of partnering with different local authorities and companies, increased coordination efforts, location-specific knowledge, etc. Importantly, these costs are usually *sunk* and take place before the actual implementation of the experiment/survey.³ As a result, the optimal depth vs width balances statistical accuracy and implementation costs.

The relevant statistical component entering the design problem is the intra-cluster correlation (ICC), i.e. how much individuals vary within their cluster compared to how much they vary overall. In particular, the marginal value of an additional observation from the same cluster (in terms of variance reduction) is lower whenever the ICC is higher. As a result, the designer would prefer wide designs (many clusters, few units per cluster) in the presence of high ICC levels, and would prefer deep designs (few clusters, many units) in the presence of low ICC levels.⁴ While the cost structure is usually known to the designer, it is often difficult to make strong assumptions on the share of ICC, especially given its large variability across regions and domains. For reference, in the context of medical studies, [Knox and Chondros \(2004\)](#) encounter ICC estimates varying an order of magnitude across covariates within the same population, thus the empirical spread of ICC coefficients is indeed a real phenomenon.⁵ The variability of the ICC and the design’s sensitivity to this parameter—see Figure 1—make its imputation a first order concern for experimentalists.

While the ICC is empirically relevant, its estimation may be quite challenging. To see why, observe that the ICC can be written as a function of the within- and between-cluster variances. The estimation of the latter requires sampling from many clusters, which is, precisely, the expensive element of the design. This dynamic places the designer at a conundrum: Optimal design identification requires knowing the ICC but ICC estimation requires sampling from many clusters. Because cluster exploration often requires bearing fixed costs that cannot be undone, ICC estimation damages optimal design implementation.

To the author’s knowledge, this is the first paper to investigate the depth-vs-width problem

²Some additional examples include: surveys to CEOs across different industries [Datta and Rajagopalan \(1998\)](#), land surveys across different regions [Lin and Ho \(2003\)](#), treatment allocation to public clinics across different districts [Karing \(2024\)](#), treatment allocation to villages across different wards [Heß et al. \(2021\)](#), and treatment allocation to districts across different provinces [Ashraf et al. \(2020\)](#).

³For instance, [Meriggi et al. \(2024\)](#) need to train different social-mobilization teams in rural Sierra Leone, and [Li et al. \(2026\)](#) need to build an algorithm around the hiring infrastructure of a big company. These costs are nearly invariable to the number of treated units in each region/company, and must be covered before the actual implementation of the experiment.

⁴In the extreme case where observations are perfectly correlated within clusters, the analyst extracts all the cluster’s information by sampling a single individual. In this extreme setting, the analyst would sample a single unit from as many clusters as the budget allows for.

⁵[Knox and Chondros \(2004\)](#) encounter ICC estimates ranging from 0.055 to 0.451 on demographic covariates, and from 0.005 to 0.059 on morbidity covariates, both set of covariates showing an order of magnitude variation.

with unknown intra-cluster correlation. To do so, I consider a minimal model with nested errors, linear costs and a hard budget. I then characterize the oracle policy (the variance minimizing design) by solving the designer’s problem under known ICC. Secondly, I develop two algorithms which solve the problem efficiently without any previous knowledge of the ICC. Both algorithms follow the same structure: they first run a pilot where they learn the ICC, and then, they allocate the remaining budget between units and clusters according to plug-in rules. The first algorithm implements the design in just two-stages by taking a worst-case approach over the pilot. I refer to this algorithm as Explore-then-Commit Cluster Selection (**EtC-CS**). The second algorithm implements an adaptive pilot by sequentially sampling clusters until it stops according to a data-dependent rule and commits to the final design. I refer to this algorithm as Adaptive Cluster Selection (**ACS**).

In terms of performance, I first show that the variance of the oracle design—the variance minimizing solution to the problem under known ICC—decreases linearly with the budget B , i.e. it is $O(1/B)$. I then show that both algorithms implement designs which induce negligible excess variance on the estimators compared to the optimal design. In particular, they achieve worst-case expected excess variance of order $O(\ln B/B^2)$, much smaller than the order of the variance itself. Although both algorithms display bounds of the same order, empirically, **ACS** outperforms **EtC-CS** at the expense of larger adaptability requirements.

Even if our algorithms induce estimators with low variance conditional on the design, the reader may fear that by collecting data adaptively, we might be compromising inference on the resulting estimators (Wager (2024)). Under a symmetry assumption on the errors—similar to the ones in randomization inference—we show that this is not the case. In fact, we prove a very strong result which is of independent interest for the literature on adaptive data collection: inference on sample averages is invariant to adaptive data collection procedures which only depend on sample variances. This means that asymptotic normality is preserved without further correction nor data wastage. This result hinges on a simple heuristic: under symmetry, error averages conditional on error variances are mean-zero. Therefore, by proving a stable CLT on the averages (over the process generated by the variance data), one may transform zero covariance across the errors into independence between the average and design. Both **EtC-CS** and **ACS** are functions of (sequences of) ICC estimators, which are functions of the within- and between-cluster variance, thus, theory applies.

Finally, I show that the excess variance of **EtC-CS** and **ACS** is minmax optimal for large B by deriving matching lower bounds (up to logarithmic terms) of $O(1/B^2)$. I also argue that their excess variance is non-trivial, at least when compared to canonical static policies. For reference, any static policy that takes a misspecified correlation as an input (including the minmax optimal static policy) will incur excess variance of $O(1/B)$, that is of the same order than the oracle variance and much larger than the excess variance of the proposed algorithms. A large array of simulations confirm the improved performance of our algorithms compared to static policies, the superior performance of **ACS** compared to **EtC-CS** and the design conditional asymptotic normality of the induced estimators.

Empirical Application - TBC

Extensions - TBC

The rest of the manuscript is organized as follows, Section 2 provides a brief literature review on survey theory and adaptive experimental design, Section 3 introduces the cost-aware variance

minimization problem under known ICC, Section 4 discusses estimation in the context of unknown ICC and presents the two candidate algorithms, Section 5 provides high-probability and expected excess variance upper bounds for our algorithms, Section 6 discusses inference under adaptive data collection based on sample variances, and shows asymptotic independence between the sample average estimator and the design, Section 7 derives matching lower bounds on the problem and compares the performance of our algorithms to (misspecified) static policies, Section 8 collects simulation evidence confirming performance and inference guarantees, and Section 9 concludes. All proofs can be found in the Appendix.

2 Literature Review

Survey Theory. The problem of optimal multi-stage survey design under a budget is a well-studied problem in the field of survey theory dating back to the seminal work of Kish (1965).⁶ On its simplest form, a multi-stage survey is deployed by first splitting the population of interest in a set of (non-overlapping) clusters, then, randomly selecting a number of clusters K (proportionally to their size), and, finally, randomly sampling N units from each of the selected clusters. The process may be towered by first drawing K_1 super-clusters, then K_2 sub-clusters, etc. The goal in this literature is very close to the spirit of this paper: to minimize the variance of the resulting population estimators by careful definition and/or selection of clusters and surveyed units. The literature can be divided into three broad categories: (i) sampling designs that take the clusters as given and optimize for unit selection, (ii) sampling designs that optimize cluster definition, and (iii) sampling designs that jointly optimize for cluster definition and unit selection (Ballin and Barcaroli (2013)). Our manuscript does take cluster partitions as given, thus we discuss the first category in further detail.

Stratified Survey Design. This sub-literature finds its most prominent representative on Bethel’s algorithm (Bethel (1989)), an extension of Neyman’s allocation to budget constraints and generic convex costs (Tschprow (1923), Neyman (1992)). Since the popularization of Bethel’s algorithm, the literature has focused on extending its logic to multi-domain, multi-variable and multi-objective settings, with a particular focus on algorithmic performance (Barcaroli et al. (2022)). Surprisingly, despite (i) the field’s impressive technical machinery and (ii) its emphasis on the role of intra-cluster correlation for optimal design,⁷ to the author’s knowledge there has been no attempt at introducing ICC estimation as part of the design optimization problem. Recent treatments as Barcaroli et al. (2022) assume that the cluster correlation can always be estimated from auxiliary data. In practice, the design selection seems driven by *feel*, broad logistic considerations and common misconceptions. To name a few, the UN Handbook of Household Sample Surveys in Developing and Transition Economies Tam (2005) recommends to always sample as many clusters as logistically feasible. Other sources like Sedgwick (2015) argue that the number of clusters must be fixed in advance. I will argue that none of these considerations are necessarily optimal.

The Econometrics of Stratified Experiments. My work is closely related in spirit to the econometric literature on optimal stratification and optimal covariate adjustment, albeit the designs in this literature are not really adaptive. The goal of this literature is to minimize the variance of an ATE

⁶see Biemer and Lyberg (2003) for a more recent textbook treatment, and Sedgwick (2015) for a brief discussion and policy guide.

⁷See for instance Cicchitelli et al. (1992) for an extended discussion on the role of ICC and the inclusion of several nested clusters. Knox and Chondros (2004) presents compelling empirical evidence on the variability of the ICC across domains. Campbell et al. (2004) collect a methodological note on the usage and reporting of ICC estimates.

estimator. Some of it even considers the role of sampling costs. Some examples include [Cytrynbaum \(2021\)](#), [Cytrynbaum \(2024\)](#), [Bai \(2022\)](#) or [Bai et al. \(2024\)](#). Despite being closely related theme-wise, this paper does not look into the problem of strata definition nor covariate adjustment, but at the trade-off between cluster and unit selection under a budget. Moreover, on its simplest form, the proposed solution does not rely on external covariates for its implementation.

Adaptive Experimental Design. Formally, my work resonates with the adaptive experimental design and online policy learning literature. For instance, in the context of best arm identification, [Adusumilli \(2026\)](#) considers the problem of optimal stopping in sampling experiments, [Imbens et al. \(2025\)](#) shows the inadmissibility of uniform sampling rules in multi-arm settings, and [Kasy and Sautmann \(2021\)](#) discuss sequential treatment allocation. We further encounter theoretical and applied papers on bandit learning in Economics with a similar notion of regret to the one described in this paper such as [Cesa-Bianchi et al. \(2025\)](#) and [Li et al. \(2026\)](#).

Inference on Adaptive Designs. The literature on inference with adaptive designs is relatively thin and full of negative results: In general, adaptive data selection compromises estimation and invalidates inference. [Nie et al. \(2018\)](#) argue that most adaptive data collection procedures deliver negatively biased sample average estimates. Similarly, [Hadad et al. \(2021\)](#) show that IPW estimators are not asymptotically normal and suffer from fat tails.⁸ Unlike most data collection procedures in this literature, our algorithms do not depend on the levels but on the second moments of the data. This difference allows us to escape from the malaise of adaptive data collection (at least under homogeneous clusters with balanced designs). Asymptotic independence is established under an error symmetry assumption, which resembles results in randomization inference as the ones in [Canay et al. \(2017\)](#).

Others. Finally, a shallow connection exists between my work and the bandit literature on optimal sampling designs, going back to the equivalence result of D-optimal and G-optimal problems in [Kiefer and Wolfowitz \(1960\)](#).⁹ Nonetheless, the tools and scope in this literature are quite removed from the analysis in this manuscript.

3 Design Problem under Known ICC

3.1 Survey Design Problem

Consider a population with $\bar{K}$ equally sized clusters containing $\bar{N}$ units each. The goal of the researcher is to minimize the MSE of an estimator of a population average by deciding how many units $N_k \ll \bar{N}$ sample for each cluster, where $K = \sum \mathbb{1}(N_k > 0) \ll \bar{K}$ is the number of sampled clusters. Clusters are indexed by $k = 1, \dots, K$ and units are indexed by $i = 1, \dots, N_k$ for each cluster k . Let y_{ki} be the variable of interest which follows an additive model with nested errors,¹⁰

$$y_{ki} = \mu + u_k + \varepsilon_{ki}, \tag{1}$$

⁸See [Wager \(2024\)](#) for a recent textbook treatment.

⁹See [Lattimore and Szepesvári \(2020\)](#) for a more recent textbook treatment.

¹⁰The variance minimization problem and excess variance analysis only rely on certain moment restrictions implied by Equation 1. In that sense, our analysis extends beyond this additive model. However, the additive error structure of the model plays an important role in our inference analysis, thus, for internal consistency and simplicity, we take Equation 1 as the model of reference.

where y_{ki} is the variable of interest, u_k is the cluster-level error (*iid* with mean 0 and variance ρ), ε_{ik} is the individual-level error (also *iid* with mean 0 and variance σ), and $u_k \perp\!\!\!\perp \varepsilon_{ki}$ for all k, i . Define $\lambda = \sigma/\rho$ as the **variance ratio**. λ is the main object of interest entering the analysis, and is connected one-to-one to the intra-cluster correlation, $\text{ICC} = \rho/(\rho + \sigma) = 1/(1 + \lambda)$. The model in Equation 1 implicitly assumes the following moment restrictions: (i) cross-cluster independence, i.e. $\text{CoVar}(y_{ki}, y_{lj}) = 0$ for all i, j and $k \neq l$; (ii) homogeneous total variance, i.e. $\text{Var}(y_{ki}) = \rho(1 + \lambda)$ for all k, i ; and (iii) within-cluster equicorrelation, i.e. $\text{CoVar}(y_{ki}, y_{kj}) = \rho$ for all i, j , and all k . Let $\omega = (\rho, \lambda)$ such that

Assumption 3.1. *Compact Parameter Space.* ω lives in a known compact set $\Omega = [\rho_m, \rho_M] \times [\lambda_m, \lambda_M]$ where $[\rho_m, \rho_M]$ is a strict subset of $(0, \infty)$.

For simplicity in the exposition, we focus on the case where $N_k \in \{0, N\}$ for all k . In the homogeneous environment setting this is without loss, as the variance minimizing allocation is indeed a balanced one.¹¹ The general setting is discussed in Appendix A. For fixed (K, N) , the natural estimator of the population parameter μ is simply the **average of averages**,

$$\hat{\mu} = \hat{\mu}(K, N) = \frac{1}{KN} \sum_{k=1}^K \sum_{i=1}^N y_{ki}$$

which is unbiased for μ under multi-stage randomization (and mean-zero noise).

3.2 Experimental Design

The same design problem arises in a simple cluster-stratified experiment. Suppose the researcher samples an even number of clusters $K \ll \bar{K}$ with $N \ll \bar{N}$ units per cluster. She then randomly assigns treatment at the cluster level to $K/2$ clusters. Let $D_k \in \{0, 1\}$ denote the treatment status of cluster k . Using standard potential-outcomes notation, let $Y_{ki}(1)$ and $Y_{ki}(0)$ denote the treated and untreated potential outcomes of unit i in cluster k , and define the observed outcome by

$$Y_{ki} = D_{ki}Y_{ki}(1) + (1 - D_{ki})Y_{ki}(0)$$

The object of interest is the average treatment effect

$$\text{ATE} = \mathbb{E}[Y_{ki}(1) - Y_{ki}(0)] = \mu_1 - \mu_0$$

To parallel the survey framework, assume that each potential-outcome process follows the same nested-error structure as in Equation 1, i.e. $Y_{ki}(d) = \mu_d + u_k^d + \varepsilon_{ki}^d$ for $d \in \{0, 1\}$, with $u_k^d \perp\!\!\!\perp \varepsilon_{ki}^d$.¹² Given the balanced randomization within each sampled cluster, the natural estimator of the average treatment effect is the **differences in means** estimator,

$$\hat{\text{ATE}}(K, N) = \frac{2}{K} \left(\sum_{k:D_k=1}^K \frac{1}{N} \sum_{i=1}^N y_{ki} - \sum_{k:D_k=0}^K \frac{1}{N} \sum_{i=1}^N y_{ki} \right)$$

¹¹The lower bounds in Section 7 do not restrict to policies that only sample balanced allocations. Therefore, our balanced allocation algorithms are near-optimal in a very general sense.

¹²Clearly, independence may not hold across treatment status, i.e. $(u_k^0, \varepsilon_{ki}^0) \not\perp\!\!\!\perp (u_k^1, \varepsilon_{ki}^1)$.

Under multi-stage randomization and mean-zero noise, $\hat{\text{ATE}}(K, N)$ is unbiased for the ATE.

3.3 Variance Minimization under Linear Costs

The problem is completely symmetric in the survey and experimental design settings, so we focus on the survey setting as the operating framework. In both settings, the goal of the learner is to minimize the MSE of the estimator $\hat{\mu}$ subject to linear costs and a hard budget B . Under two-stage randomization, $\hat{\mu}$ is unbiased for μ , hence, the problem reduces to a constrained variance minimization problem

$$\min_{K \geq 2, N \geq 2} W(K, N, \omega) = \frac{1}{K} \rho \left(1 + \frac{\lambda}{N} \right) \quad \text{s.t.} \quad FK + VKN \leq B \quad (2)$$

where $W(K, N, \omega) = \text{Var}(\hat{\mu}(K, N))$, F is the fixed cost of sampling an additional cluster, V is the variable cost of sampling an additional unit and $B \geq 2(F + 2V)$ is the hard budget.¹³ The equality follows from simple algebra under cross-cluster independence. The variance for $\hat{\text{ATE}}$ in the experimental design context is exactly the same. From now on, we leave integer considerations aside and focus on the **continuous relaxation** of the problem in Equation 2.¹⁴

A first immediate result is that for any $\rho > 0$, $\hat{\mu}(K, N)$ is consistent for μ if and only if $K \rightarrow \infty$. Intuitively, large N can only reduce the individual level noise ε_{ik} , while large K is necessary to drive the cluster level noise u_k to 0. Finally, given the linearity of the cost structure, the problem admits a representation as an *unconstrained* minimization problem over a single design variable N . Solving from the budget constraint

$$K^*(N) = \frac{B}{F + VN}, \quad \min_{2 \leq N \leq (B-2F)/(2V)} \frac{F + VN}{B} \rho \cdot \left(1 + \frac{\lambda}{N} \right)$$

We can solve for the optimal design $(K^*(\omega), N^*(\omega))$ by using first order conditions, i.e.

$$K^*(\omega) = \frac{B}{F + VN^*(\omega)}, \quad N^*(\omega) = \min \left\{ \max \left\{ 2, \sqrt{\lambda \frac{F}{V}} \right\}, \frac{B - 2F}{2V} \right\} \quad (3)$$

We note immediately that the map $(K^*(\omega), N^*(\omega))$ depends on ω only through the variance ratio λ . Accordingly, define the **optimal map** as $\lambda \mapsto (K^*(\lambda), N^*(\lambda))$. Despite its name, the optimal map is only really *optimal* if the designer takes the true λ as input. In general, applying the optimal map to a misspecified λ can have sizable implications on the design and the variance of the estimator induced by this design. Figure 1 plots $(K^*(\lambda), N^*(\lambda))$ and the excess variance of the resulting estimator given the design $(K = K^*(\lambda), N = N^*(\lambda))$ for different values of λ compared to the oracle policy $(K = K^*(\lambda_0), N = N^*(\lambda_0))$. Figure 1 confirms that small λ misspecification can have large implications on the induced design and the resulting excess variance.¹⁵

¹³The lower bound $K \geq 2, N \geq 2$ has been imposed for technical reasons that affect the estimation of the parameters in Section 4. In practice, this restriction is with little loss as *all* clustered surveys/experiments consider larger samples anyway.

¹⁴Although the theory is derived by disregarding integer considerations, effective algorithm implementation is integer constrained. All throughout the text, including Tables and Graphs, final integer allocation has been produced by considering the floor of (K, N) . This procedure has been applied to own and competitor algorithms to level the field and ease cross-algorithms comparison. Replication code can be found in the Supplementary Material.

¹⁵See Section 8 for further discussion.

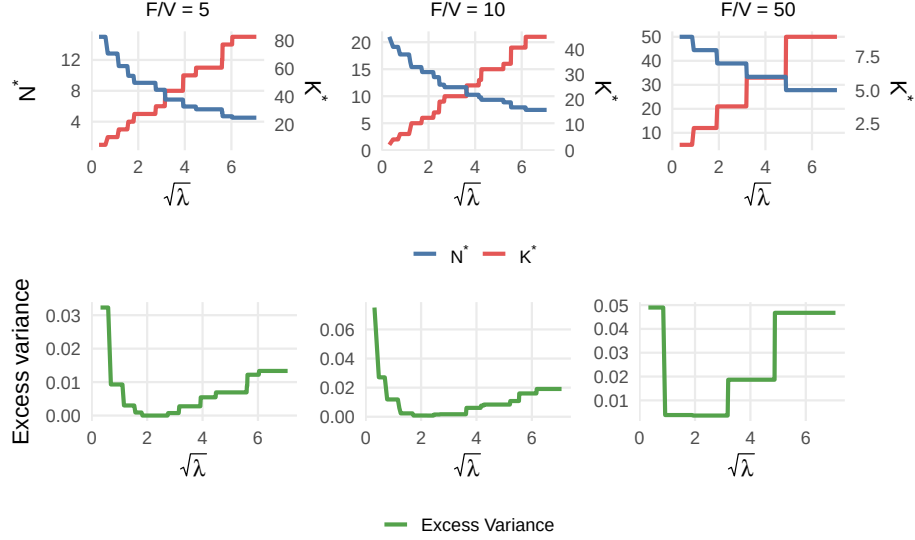


Figure 1: $N^*(\lambda)$ (top-left), $K^*(\lambda)$ (top-right) and Excess Variance (bottom) for different F/V ratios. $\mu = 0$, $\omega_0(1, 5)$, $B = 1000$, $V = 2$

The variance of the oracle policy—in the continuous relaxation of the problem—that has access to true $\lambda = \lambda_0$ is given by

$$W(K^*(\lambda_0), N^*(\lambda_0), \omega_0) = \frac{\rho_0 \left(\sqrt{V\lambda_0} + \sqrt{F} \right)^2}{B}$$

Any sampling policy which does not know λ_0 will find in the **oracle variance** a natural benchmark to compete against. Finally, we include one additional assumption to avoid dealing with (uninteresting) corner solutions.

Assumption 3.2. Interior Solution. *Given a compact set Ω such that $\lambda \in [\lambda_m, \lambda_M]$, $\lfloor N^*(\lambda_m) \rfloor \geq 2$ and $\lceil N^*(\lambda_M) \rceil \leq \frac{B-2F}{2V}$, with $B \geq 2(F + 2V)$.*

Assumption 3.2 can also be interpreted as the minimally viable experiment $N = 2, K = 2$ being compatible with the parameter space.¹⁶

4 Design Problem under Unknown ICC

4.1 Estimation of λ

Optimal survey design under unknown model parameters will depend on some form of parameter estimation, therefore, we start this section by characterizing the relevant estimators. Fix (K, N) with

¹⁶Assumption 3.2 uses the fact that N^* is increasing in λ . Moreover, the condition $\lceil N^*(\lambda_M) \rceil \leq (B - 2F)/2V$ can be equally stated as $\lfloor K^*(\lambda_M) \rfloor \geq 2$.

$K \geq 2, N \geq 2$ and define

$$\begin{aligned}\hat{\mu}^k(N) &= \frac{1}{N} \sum_{i=1}^N y_{ki} & \tilde{w}(K, N) &= \frac{1}{K(N-1)} \sum_{k=1}^K \sum_{i=1}^N (y_{ki} - \hat{\mu}^k(N))^2 \\ \tilde{b}(K, N) &= \frac{1}{K-1} \sum_{k=1}^K (\hat{\mu}^k(N) - \hat{\mu}(K, N))^2 & \tilde{\rho}(K, N) &= \tilde{b}(K, N) - \frac{\tilde{w}(K, N)}{N}\end{aligned}$$

where $\hat{\mu}^k$ is the sample average for cluster k , $\tilde{w}$ is the within-cluster sample variance, $\tilde{b}$ is the between-cluster sample variance and $\tilde{\rho}$ is sample estimator for ρ . Now define the clip operator $\Pi_{[a,b]}(x) = \min\{b, \max\{a, x\}\}$ and consider the clipped analogues

$$\hat{w}(K, N) = \Pi_{[\lambda_m \rho_m, \lambda_M \rho_M]}(\tilde{w}(K, N)) \quad \hat{\rho}(K, N) = \Pi_{[\rho_m, \rho_M]}(\tilde{\rho}(K, N)) \quad \hat{\lambda}(K, N) = \frac{\hat{w}(K, N)}{\hat{\rho}(K, N)}$$

where $\hat{\lambda}(K, N)$ is the (clipped) sample estimator of λ . Lemma 4.1 shows consistency of these estimators.¹⁷

Lemma 4.1. *Estimator Properties.* Let $\xrightarrow[T]{P}$ note convergence in probability under T asymptotics and assume $\omega_0 \in \Omega$, then (i) $\hat{w}(K, N) \xrightarrow[N]{P} \sigma_0$ (with fixed $K \geq 1$) and $\hat{w}(K, N) \xrightarrow[K]{P} \sigma_0$ (with fixed $N \geq 2$); (ii) $\hat{\rho}(K, N) \xrightarrow[K]{P} \rho_0$ (with fixed $N \geq 2$); and (iii) $\hat{\lambda}(K, N) \xrightarrow[K]{P} \lambda_0$ (with fixed $N \geq 2$)

There are two important takeaways from the definitions in Equation 4.1 and Lemma 4.1. Firstly, the variance ratio λ is a function of the within- and between-cluster variances. Secondly, while the within-variance can be estimated consistently under both N and K asymptotics, the between-variance (and therefore ρ) can only be consistently estimated under K asymptotics. As a result, only K asymptotics deliver consistent estimation of λ . An important intuition for the construction of algorithms follows

Corollary 4.2. *Consistent Pilots.* Consider a policy π that implements

$$(K^\pi, N^\pi) = (K^*(\hat{\lambda}(K_p, N_p), N^*(\hat{\lambda}(K_p, N_p)))$$

for some pilot design with random size $(K_p(B), N_p(B))$ such that $\lim_{B \rightarrow \infty} K_p(B) = \infty$ a.s., then, $(K^\pi, N^\pi) \xrightarrow[B]{P} (K^*(\lambda_0), N^*(\lambda_0))$.

Remark on Corollary 4.2. The proof of a stronger version for the Corollary can be found in the Appendix (as part of Lemma C.3). The Corollary suggest that any policy which first runs a pilot to learn λ and then uses plug-in rules under the optimal map, must sample an asymptotically large number of clusters during the pilot.

4.2 Explore-then-Commit Cluster Sampling

In this section we propose a simple solution to the problem of the designer under unknown ICC using an Explore-then-Commit structure. As anticipated in Corollary 4.2, consider a sampling policy which first conducts a pilot with size (K_p, N_p) , then constructs an estimator of the form $\hat{\lambda}_p = \hat{\lambda}(K_p, N_p)$, and,

¹⁷For simplicity in asymptotic analysis, we will some times consider the uncorrected version of the estimators above. This simplification is without order-loss as the uncorrected estimators remain consistent and the variation in performance/inference is order-negligible compared to the corrected estimators.

finally, *implements* the optimal map $(K^*(\hat{\lambda}_p), N^*(\hat{\lambda}_p))$. This intuitive protocol, however, presents a major flaw: There is no guarantee that $(K^*(\hat{\lambda}_p), N^*(\hat{\lambda}_p))$ is budget implementable, as the learner could have oversampled clusters or observations compared to the predicted oracle design. More precisely, the predicted optimal map is implementable iff $K_p \leq K^*(\hat{\lambda}_p)$ and $N_p \leq N^*(\hat{\lambda}_p)$.

Under Assumptions 3.1 and 3.2, a safe way to guarantee the implementability of the predicted optimal map is setting $K_p = K^*(\lambda_M)$ and $N_p = N^*(\lambda_m)$. Then, by definition of the clipped estimator, for any $\hat{\lambda}_p$, it holds that $K_p \leq K^*(\hat{\lambda}_p)$ and $N_p \leq N^*(\hat{\lambda}_p)$.¹⁸ We refer to this two-stage protocol as Explore-then-Commit Cluster Selection (**EtC-CS**). Algorithm 1 presents pseudo-code for this sampling protocol.

Algorithm 1 EtC-CS

Input $B, F, V, \lambda_m, \lambda_M$

Sample $K_p = K^*(\lambda_M)$ clusters and $N_p = N^*(\lambda_m)$ units per cluster

Recover $\hat{\lambda}_p = \hat{\lambda}(K_p, N_p)$

Implement $K^*(\hat{\lambda}_p)$ and $N^*(\hat{\lambda}_p)$

4.3 Adaptive Cluster Selection

While theorists emphasize the superior performance of adaptive designs, practitioners often encounter logistical challenges that prevent their full implementation (Chow and Corey (2011)). In that sense, we praise the efficacy and simplicity of **EtC-CS** which achieves near-optimal performance guarantees (as per Theorem 5.2) with minimal adaptability—just a two-stage design. However, the reader may wonder what can be achieved if the designer was given access to further adaptability. Consider the following thought experiment. A researcher must select a design that samples something between 20 and 60 clusters, i.e. given a budget B , $K^*(\lambda_M) = 20$ and $K^*(\lambda_m) = 60$. **EtC-CS** would then recommend sampling 20 clusters as part of the pilot, compute $\hat{\lambda}_p = \hat{\lambda}(20, N_p)$, and implement $(K^*(\hat{\lambda}_p), N^*(\hat{\lambda}_p))$. Now, imagine that the design prescribed by the pilot is $K^*(\hat{\lambda}_p) = 50$. It would seem unreasonable to commit straightaway to this design.¹⁹ In practice, the designer could safely explore a few more clusters to gather more information on λ before committing to the final design, with minimal risk of incurring cluster oversampling. Based on this heuristic, we propose an improved algorithm which samples clusters sequentially until it stops according to a data-dependent rule and implements the final design. We refer to this algorithm as Adaptive Cluster Selection (**ACS**). The size of the pilot in **ACS** is determined by the stopping time τ , such that

$$\tau = \inf\{k : k + 1 > K^*(\hat{\lambda}(k, N_p))\} \wedge \left\lfloor \frac{B}{F + VN_p} \right\rfloor \quad (4)$$

In other words, the adaptive algorithm explores N_p units from an additional cluster until sampling the new cluster would incur oversampling according to the online estimator of the optimal design. By treating N_p as fixed, and exploring only in the cluster dimension, this heuristic sampling process combines simplicity with statistical guarantees (recall Corollary 4.2). In principle, under a similar heuristic, the learner may consider sampling a new unit—either from an already explored clus-

¹⁸Observe that, in some sense, the proposed design is minmax optimal for a learner that wants to implement a second stage design via optimal mapping, i.e. $(K^*(\hat{\lambda}_p), N^*(\hat{\lambda}_p))$ for some consistent $\hat{\lambda}$.

¹⁹Especially, if the learner could reject a null of the form $H_0 : K^*(\lambda_0) < \tilde{K}$ for some $\tilde{K} \in [21, 49]$

ter or from a new cluster—to maximize the (cost adjusted) expected marginal return to exploration. However, this allocation process is not only computationally infeasible but also highly impractical in real applications. In addition, as per the discussion earlier in this Section, reducing the exploration to the K dimension is the correct simplification. In small samples, sampling additional units from already explored clusters may reduce the variance of the $\hat{\lambda}$ estimator. However, the asymptotic variance is independent of N and only strictly decreasing in K . It follows that, when it comes to λ estimation, asymptotically, the learner has an incentive to commit all her exploration resources to cluster exploration. Algorithm 2 formalizes this sampling protocol

Algorithm 2 ACS

Input $B, F, V, \lambda_m, \lambda_M$
Initialize $K = 0, N_p = N^*(\lambda_m), \hat{\lambda}_0 = \lambda_m$
while $K + 1 \leq \min \left\{ K^*(\hat{\lambda}_K), \left\lfloor \frac{B}{F+VN_p} \right\rfloor \right\}$
 explore an additional cluster k and **observe** $(y_{ki})_{i=1}^{N_p}$
 update $K = K + 1$
recover $\hat{\lambda}_{\text{ACS}} = \hat{\lambda}(K - 1, N_p)$ using $((y_{ki})_{i=1}^{N_p})_{k=1}^{K-1}$
implement $K^*(\hat{\lambda}_{\text{ACS}})$ and $N^*(\hat{\lambda}_{\text{ACS}})$

Beyond adaptability, the key distinction between Algorithms 1 and 2 is that ACS concentrates all cluster exploration during the pilot, while EtC-CS explores new clusters after the pilot. This allows ACS to use as many clusters as possible for λ estimation. Recall that the estimation of λ improves with K , which, incidentally, is the expensive element of the design. Hence, there is no reason to waste information for λ estimation by excluding clusters.²⁰ The use of information from $\tau - 1$ clusters for $\hat{\lambda}$ estimation is a minor technical concern. Given a stopping time τ it must have been the case that $\tau \leq K^*(\hat{\lambda}_{\tau-1})$, otherwise the algorithm would have not explored cluster τ . Moreover, $\tau + 1 > K^*(\hat{\lambda}_\tau)$. However, from these two facts alone one cannot infer the natural condition $\tau \leq K^*(\hat{\lambda}_\tau)$, but only $\tau \leq K^*(\hat{\lambda}_{\tau-1})$. It follows that the design $(K^*(\hat{\lambda}_{\tau-1}), N^*(\hat{\lambda}_{\tau-1}))$ is implementable a.s. under Assumption 3.2.

5 Upper Bounds on Excess Variance

This Section presents the main results in terms of algorithmic performance. In particular, we upper bound the excess variance for each algorithm under a benign high-probability event (Theorem 5.2). We then derive an unconditional worst-case excess variance for any algorithm (Lemma 5.3), and use this result to transform the high-probability bounds into expected excess variance bounds (Theorem 5.4). These bounds are of order $O(\ln B/B^2)$ for both algorithms.

For any policy π , define the excess variance of the policy as the random variable governing the difference between the variance of the design (K^π, N^π) induced by policy π and the variance of the oracle policy, i.e.

$$\Delta_\pi = W(K^\pi, N^\pi, \omega_0) - W(K^*(\lambda_0), N^*(\lambda_0), \omega_0)$$

²⁰This dynamic can also be modeled in terms of incentives between a λ learner and a maximally risk averse designer that wants to implement a design of the form $(K^*(\hat{\lambda}_p), N^*(\hat{\lambda}_p))$ for some pilot $\hat{\lambda}_p$. During the first $K^*(\lambda_0)$ clusters, the preferences of both individuals are perfectly aligned.

We now impose a (benign) subgaussian assumption on the distribution of errors. The subgaussian assumption is quite standard in the online learning and policy learning literature. It essentially enforces sufficient concentration of the error terms. Similar derivations will go through under a weaker subexponential assumption.

Assumption 5.1. *Errors Subgaussianity.* Let u_k be $\sqrt{\kappa_u}$ -subgaussian and ε_{ki} be $\sqrt{\kappa_\varepsilon}$ -subgaussian.

Theorem 5.2. *High Probability Bounds on EtC-CS and ACS.* Under Assumptions 3.1, 3.2 and 5.1, with probability at least $1 - \alpha$, there exists constants $C_1, C_2 < \infty$ independent of B such that

$$\Delta_{\text{EtC-CS}} \leq C_1 \frac{\ln(1/\alpha)}{B^2} \quad \Delta_{\text{ACS}} \leq C_2 \frac{\ln(1/\alpha)}{B^2}$$

Interpretation of Theorem 5.2. Proof of Theorem 5.2 can be found in Appendix C. The main takeaway of Theorem 5.2 is that, with high probability, the excess variances of EtC-CS and ACS are of order $O(1/B^2)$, therefore, both of these algorithms induce asymptotically negligible excess variance to the oracle variance—which is of order $O(1/B)$. In other words, a designer that implements a design according to EtC-CS or ACS is in high-probability not worse (asymptotically) than an oracle designer who knows λ ex-ante. We now explain the mechanics and intuition behind this result, and, in doing so, we argue that, despite sharing the same order of excess variance, ACS is likely to outperform EtC-CS. From our discussion above, both algorithms implement a near-optimal design as prescribed by some pilot-estimated $\hat{\lambda}$. This is $\hat{\lambda}_p = \hat{\lambda}(K_M, N_m)$ in the case of EtC-CS and $\hat{\lambda}_p = \hat{\lambda}(\tau - 1, N_p)$ in the case of ACS, with $\tau - 1 \geq K_m$ (under the usual assumptions). In both cases, the deviations with respect to the oracle design are local to λ , therefore we expect deviations to be smaller (in high probability/in expectation) under ACS. Yet, the increase in precision does not affect the order of the upper bounds because, even under maximal deviations, i.e. $K_p^{\text{EtC-CS}} = K_M$ and $K_p^{\text{ACS}} = K_m$, provided $K_m/K_M = O(1)$ in budget.

In fact, it is surprisingly more challenging to recover the desired order for the upper bound in the adaptive case as one needs to upper bound the deviations for a random pilot size τ . We circumvent this issue by (i) noting that $\hat{\lambda} - \lambda$ is, asymptotically, a function of martingales (centered errors), and (ii) applying a maximal version of Bernstein’s inequality for martingales. Finally, we argue that the stopping time is bounded in a window of size $O(1)$, recovering the desired rate. The expected superior performance of ACS compared to EtC-CS is verified consistently in the simulation evidence (see Section 8).

5.1 Worst Case and Expected Excess Variance

In this section we translate our high-probability upper bounds on excess variance into upper bounds on expected excess variance. To do so, we first characterize the worst case excess variance of any design which consumes the budget entirely. We then argue that with probability α the designer incurs worst case excess variance, and with probability $1 - \alpha$ excess variance of order $O(1/B^2)$ as discussed in Theorem 5.2. By careful selection of α we show upper bounds on expected excess variance of $O(\ln B/B^2)$, preserving the asymptotically negligible excess variance property of the high probability bounds, up to a logarithmic factor.

Lemma 5.3. *Worst Case Excess Variance.* Under Assumption 3.1, for any policy π such that $FK^\pi + VK^\pi N^\pi = B$ with $K^\pi \geq 2, N^\pi \geq 2$ and sufficiently large B , there exists a small positive δ such that

$$W(K^\pi, N^\pi, \omega_0) - W(K^*(\lambda_0), N^*(\lambda_0), \omega_0) \leq \frac{\rho_M}{2} + \delta \quad \forall \omega_0 \in \Omega$$

Interpretation of Lemma 5.3. As discussed in Section 3, estimator’s variance is only driven to 0 under K asymptotics. As a result, any design which samples a finite number of clusters will incur non-vanishing excess variance. In particular, the lower bound in excess variance of these policies is $\rho_0/2$. The usual compactness assumption on Ω recovers the final expression.

Theorem 5.4. *Expected Excess Variance.* Under Assumptions 3.1, 3.2 and 5.1, there exists constants $C_3, C_4 < \infty$ independent of B such that

$$\begin{aligned} \mathbb{E}_{\text{EtC-CS}}[W(K, N, \omega_0)] - W(K^*(\lambda_0), N^*(\lambda_0), \omega_0) &\leq C_3 \frac{\ln(B)}{B^2} \\ \mathbb{E}_{\text{ACS}}[W(K, N, \omega_0)] - W(K^*(\lambda_0), N^*(\lambda_0), \omega_0) &\leq C_4 \frac{\ln(B)}{B^2} \end{aligned}$$

Interpretation of Theorem 5.4. Controlling for expected excess variance is a stronger performance guarantee than controlling for excess variance with high probability. The learner can rest assured that switching across metrics does not worsen the order of excess variance beyond a $\ln(B)$ term. As a result, the excess variance of both EtC-CS and ACS remains asymptotically negligible compared to the oracle with an upper bound of $O(\ln B/B^2)$. In Section 7, we argue that the rate in the excess variance of these algorithms is not only negligible but also near-optimal—up to logarithmic terms. This means that no algorithm can substantially improve upon Algorithms 1 and 2. Finally, we show that their excess variance is also non-trivial, at least when compared to any misspecified static policy.

6 Inference

So far, we have argued vehemently in favor of the adaptive design of cluster experiments. However, the experimentalist might be concerned that by selecting the design after looking at the pilot data, the distribution of the resulting estimator conditional on the design might be compromised. Under an error symmetry assumption, we argue that this is not the case and that the distribution of the resulting estimator remains invariant after conditioning on the designs induced by EtC-CS and ACS. This means that asymptotic normality of $\hat{\mu}(K, N)$ is preserved without further correction nor pilot data exclusion. This result is of independent interest to the literature on inference with adaptive data collection.

The proof can be split in 3 steps. In the first step we argue that the designs of interest are functions of (a sequence of) $\hat{\lambda}(K, N)$ estimators. We then argue that these estimators are asymptotically even. Recall that $\hat{\lambda}(K, N)$ is a function of the within-variance and the between-variance. The within-variance is even for all (K, N) , but the between-variance is only asymptotically even, as it de-means by the grand-mean. In the second step, we define the average cluster- k errors $M_{k,N} = u_k + \frac{1}{N} \sum_{i=1}^N \varepsilon_{ki}$ (for some fixed N). We then leverage the cluster-symmetry assumption to show zero asymptotic covariance between $M_{k,N}$ and the sample variances through an odd-function even-event argument.

Assumption 6.1. *Inference Assumptions*

1. **Symmetry and Independent Errors:** Let $C_k = (u_k, \varepsilon_{k1}, \varepsilon_{k2}, \dots)$ be the vector of cluster k errors. Assume $\mathbb{E}[u_k] = \mathbb{E}[\varepsilon_{ki}] = 0$, $\text{Var}(u_k) = \rho$, $\text{Var}(\varepsilon_{ki}) = \sigma$, $C_k \stackrel{d}{=} -C_k$ with C_k iid and $u_k \perp\!\!\!\perp \varepsilon_{ki}$ for all i, k .

2. **Finite Fourth Moments:** For some small positive $\eta > 0$, $\mathbb{E}[|u_k|^{4+\eta}] < \infty$, $\mathbb{E}[|\varepsilon_{ki}|^{4+\eta}] < \infty$

This assumption is relatively common in randomization inference analyses as in [Canay et al. \(2017\)](#), yet, it remains strong. In particular, it rules out binary, count or skewed outcomes. The third step confronts the main technical challenge in the analysis. Intuitively, one would like to transform the weak convergence to independent limits of $M_{k,N}$ and the sample variances into convergence to independent limits of $\hat{\mu}(K, N) - \mu = \frac{1}{K} \sum_{k=1}^K M_{k,N}$ and the sample variances. The main problem is that the sample average function depends on the design, thus weak convergence to independent limits is not sufficient. Instead, one needs to show stable convergence. We do so by relying on a Martingale stable CLT invoking the martingale property of $M_{k,N}$. See Appendix D for further details.

Theorem 6.2. *Design Robust Inference.* Let (K^π, N^π) be the design induced by some policy $\pi \in \{\text{EtC-CS}, \text{ACS}\}$, with (K^π, N^π) $\mathcal{G}$ -measurable. Define $\theta_0 = \rho_0(1 + \lambda_0/N^*(\lambda_0))$. Then, under Assumptions 3.1, 3.2 and 6.1

$$Z_B := \frac{\sqrt{K^\pi}(\hat{\mu}(K^\pi, N^\pi) - \mu)}{\theta_0} \xrightarrow[B]{} N(0, 1) \quad \mathcal{G}\text{-stably}$$

Therefore, $\sup_x |\mathbb{P}(Z_B \leq x | (K^\pi, N^\pi)) - \Phi(x)| \xrightarrow[B]{P} 0$. The same result holds when replacing θ_0 by $\hat{b}(K^\pi, N^\pi)$, thus

$$\hat{\mu}(K^\pi, N^\pi) \pm z_{1-\alpha/2} \sqrt{\hat{b}(K^\pi, N^\pi)/K^\pi}$$

has asymptotic coverage $1 - \alpha$.

The proof of Theorem 6.2 and the definition of the relevant σ -algebra $\mathcal{G}$, can be found in Appendix D. The theorem presents a very strong result. It claims that the distribution of the induced estimator remains invariant to adaptive data collection when the collection is driven by second moments of the data, i.e. it has the same asymptotic distribution as an estimator with an ex-ante fixed design. The simulation evidence in Section 8 provides empirical support for these claims. Overall, this result opens the door for a generalized theory of inference under adaptive data collection when sampling is driven by estimates of second moments. Contrary to most results in the literature, it suggests that the researcher can preserve standard inference without further refinements nor data exclusion.

7 Lower Bound and Policy Comparison

In this Section we show that EtC-CS and ACS are essentially unimprovable by deriving lower bounds on the problem of $O(1/B^2)$, i.e. lower bounds are matching up to logarithmic terms. We then prove that any static policy which optimizes with respect to a misspecified ICC will incur excess variance of $O(1/B)$, that is, of the same order than the oracle variance and one order of magnitude larger than the excess variance of the proposed algorithms. We use this evidence to argue that the excess

variance of our algorithms is indeed non-trivial and that ICC learning is necessary for good asymptotic performance in survey and experimental designs.

7.1 Lower Bound

As anticipated, we derive lower bounds for the generalized game where the learner can select N_k arbitrarily for every k (subject to budget constraints). See Appendix A for a complete characterization. In order to derive lower bounds for this regime, we consider an *easier* version of the problem where, every time the learner samples N_k units cluster from cluster k she does not only observe $(y_{ik})_{N_k}$ realizations, but she also observes the cluster noise u_k . The problem is easier inasmuch as it contains weakly more information for the learner than the standard experiment. By deriving lower bounds on the augmented experiment we effectively guarantee lower bounds on the benchmark game. This simplification saves us some technical problems. Because observations within the same cluster are correlated, when cluster k has already been visited, the new observations from that cluster are not identically distributed conditional on the history of observations. This lack of independence invalidates standard *divergence decomposition* arguments in the original experiment. By revealing the cluster noise u_k when cluster k is explored for the first time, history-conditional independence is restored and standard proof arguments go through. The rest of proof is quite mainstream. We consider two data generating processes y_- , y_+ with the same expectation μ —indexed by ω_- and ω_+ —following the structure of Equation 1. Define

$$\omega_- = (\rho_-, \lambda_-) \quad \omega_+ = (\rho_+, \lambda_+) \quad \lambda_- = \sigma/\rho_- \quad \lambda_+ = \sigma/\rho_+$$

For some $\lambda \in (\lambda_m, \lambda_M)$ and some small positive δ , ρ_- and ρ_+ are selected such that $\lambda_- = \lambda - \delta$ and $\lambda_+ = \lambda + \delta$. In other words, the only difference across the DGPs is some small variation in the cluster variance ρ . As a result, optimizing with respect to λ_- is optimal in environment y_- and optimizing with respect to λ_+ is optimal in environment y_+ . Figure 2 illustrates this intuition.

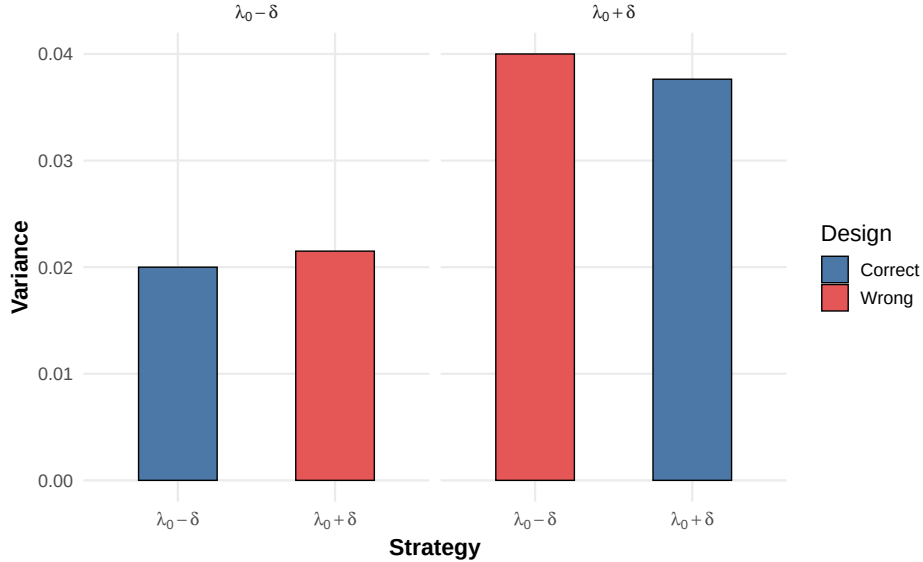


Figure 2: True Variance $W(K, N, \cdot)$ under ω_- (left) and ω_+ (right). $\mu = 0$, $\omega_0 = (1, 5 \pm \delta)$, $\delta = 0.5$, $B = 2000$, $F = 20$, $V = 2$.

δ is selected to balance a dual-objective: punishment and identification. A large δ makes it easier for the learner to distinguish across the two environments (and, therefore, to optimize accordingly), but it also penalizes her more heavily in case she fails to identify the correct instance. Conversely, a smaller δ makes it more difficult to tell instances apart, but it also makes the additional excess variance of selecting a suboptimal design smaller. By selecting $\delta \sim O(B^{-1/2})$ —as it is customary in learning problems—excess variance is maximized. Theorem 7.1 summarizes the main result in this Section. The proof of Theorem can be found in Appendix E.

Theorem 7.1. *Lower Bound on Excess Variance. For any policy $\pi \in \Pi$, there exists a DGP y and a positive finite constant $C < \infty$ such that*

$$\mathbb{E}_\pi [W(K, N, \omega_0)] - W(K^*(\lambda_0), N^*(\lambda_0), \omega_0) \geq \frac{C}{B^2}$$

Theorem 7.1 establishes a lower bound for a very general policy class Π .²¹ In particular, we allow policies to explore different number of units across different clusters, to revisit clusters freely, and to sample a different number of units every time the cluster is visited.

7.2 Policy Comparison

The results in the previous section suggest that our algorithms are unimprovable up to logarithmic terms. However, it is worth asking if the rate in the excess variance is actually trivial, in the sense that default policies would achieve a similar rate. Because this is the first paper that fits learning within survey design, we may only compare our algorithm to static design policies. In particular, we compare EtC-CS and ACS to any static policy that takes a misspecified parameter λ as input and optimizes over it, including the minmax optimal policy. By doing so, we show that the rate in the excess variance of Algorithms 1 and 2 is indeed non-trivial as all these policies accrue (expected) excess variance of $O(1/B) \gg O(\ln B/B^2)$.²²

Lemma 7.2. *Misspecified Static Policies and Minmax Optimal Static Policy. Let $\lambda_b, \lambda_0 \in [\lambda_m, \lambda_M]$ such that $\lambda_b = \lambda_0 + b$ for some $b \in \mathbb{R}$. Define the static policy $\pi_b : (K^{\pi_b}, N^{\pi_b}) = (K^*(\lambda_b), N^*(\lambda_b))$, then for some positive finite constant C*

$$W(K^{\pi_b}, N^{\pi_b}, \omega_0) - W(K^*(\lambda_0), N^*(\lambda_0), \omega_0) = C \frac{b^2}{B} + O\left(\frac{b^3}{B}\right) \sim O(1/B)$$

Moreover, define the optimal minmax static policy $\pi_{mM} : (K^{\pi_{mM}}, N^{\pi_{mM}}) = (K^*(\lambda_{mM}), N^*(\lambda_{mM}))$ where

$$\lambda_{mM} = \arg \min_{\lambda} \max_{\omega_0 \in \Omega} W(K^*(\lambda), N^*(\lambda), \omega_0) - W(K^*(\lambda), N^*(\lambda), \omega_0)$$

²¹The exact definition of the policy class Π can be found in the Appendix. The restriction to the policy class Π essentially disregards policies which do not terminate in finite time almost surely or do not satisfy the budget constrain almost surely.

²²Conditional on the design, the variance of the estimator is not random. As a result, all notions of variance (expected excess variance, excess variance in high probability, etc.) coincide for static policies that determine the exploration ex-ante.

Then,

$$W(K^{\pi_{mM}}, N^{\pi_{mM}}, \omega_0) - W(K^*(\lambda_0), N^*(\lambda_0), \omega_0) = \frac{C(\lambda_{mM} - \lambda_0)^2}{B} + O\left(\frac{(\lambda_{mM} - \lambda_0)^3}{B}\right) \sim O(1/B)$$

The Lemma above shows that any misspecified static policy (including the minmax optimal) obtains a worst-case excess variance of the same order than the variance of the oracle policy itself and much larger than the worst-case excess variance of the learning protocols $O(\ln B/B^2)$. This result is not really surprising. When there is no scope for learning, any deviation from the true λ_0 leads to a fixed second-order deviation. This preliminary result suggest that the excess variance of the policy is indeed non-trivial, at least when compared against static designs.

8 Simulations

We start this section by looking at the distribution of designs (K^π, N^π) induced by **EtC-CS** and **ACS** in Figure 3. As we can see, the distribution of designs concentrates around the oracle design $(K^*(\lambda_0), N^*(\lambda_0))$ with **ACS** overall tracking the oracle better than **EtC-CS**. This is to be expected as both algorithms induce optimal designs in the limit with **EtC-CS** using weakly less information for estimation than **ACS**. Figure 3 also plots the distribution of the estimated $\hat{\lambda}_p$ by both algorithms at the end of the exploration stage. Although the estimation is relatively sharp, crucially, it displays a long right tail. This is a small sample artifact, but it reflects a real phenomenon. Recall that $\hat{\lambda}$ is estimated as the ratio of the within variance $\hat{w}$ and the cluster variance $\hat{\rho} = \hat{b} - \hat{w}/N_p$. When ρ is small relative to σ , $\hat{\rho}$ is estimated as the small difference of two large quantities, making the overall estimation quite noisy. Because $\hat{\rho}$ enters the estimator $\hat{\lambda}$ through a (very) convex map $\hat{\rho} \mapsto 1/\hat{\rho}$, values of $\hat{\rho}$ near 0 will induce an unavoidable long right tail.

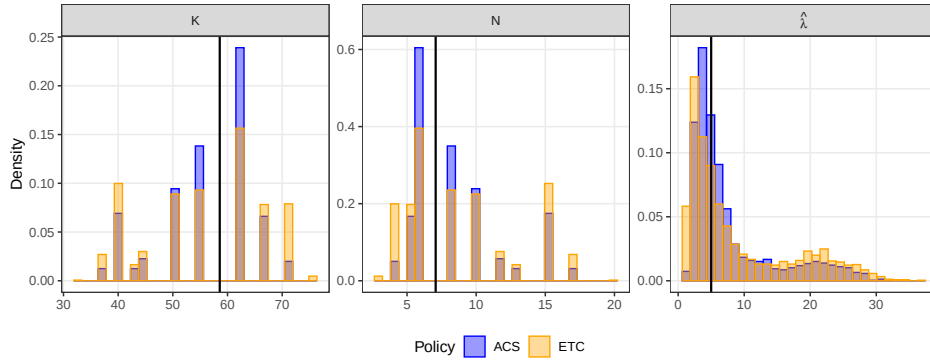


Figure 3: Design (K, N) and $\hat{\lambda}$ after Pilot Distributions induced by **EtC-CS** (yellow) and **ACS** (blue). $\mu = 0$, $\omega_0 = (1, 5)$, $B = 2000$, $F = 20$, $V = 2$. Solid black line represents oracle $K^*(\lambda_0), N^*(\lambda_0), \lambda_0$.

This long tail, however, is partially controlled by the design itself, as $\hat{\lambda}$ enters $N^*(\cdot)$ via a square-root which operates as a concave compressor. In addition, $K^*(\cdot)$ is decreasing in $\hat{\lambda}$, so, despite being convex, it is $\hat{\lambda}$ undershooting and not $\hat{\lambda}$ overshooting what could be problematic for the induced $K^*(\hat{\lambda})$. Overall, we expect some unit oversampling and cluster undersampling for large values of λ . Figure 4 confirms this pattern. Overall, both algorithms track the optimal design relatively well with **ACS** outperforming **EtC-CS**. The superior performance is more salient in the median design compared

to the average design. This is also to be expected, as ACS is, by definition, a more volatile policy than EtC-CS.

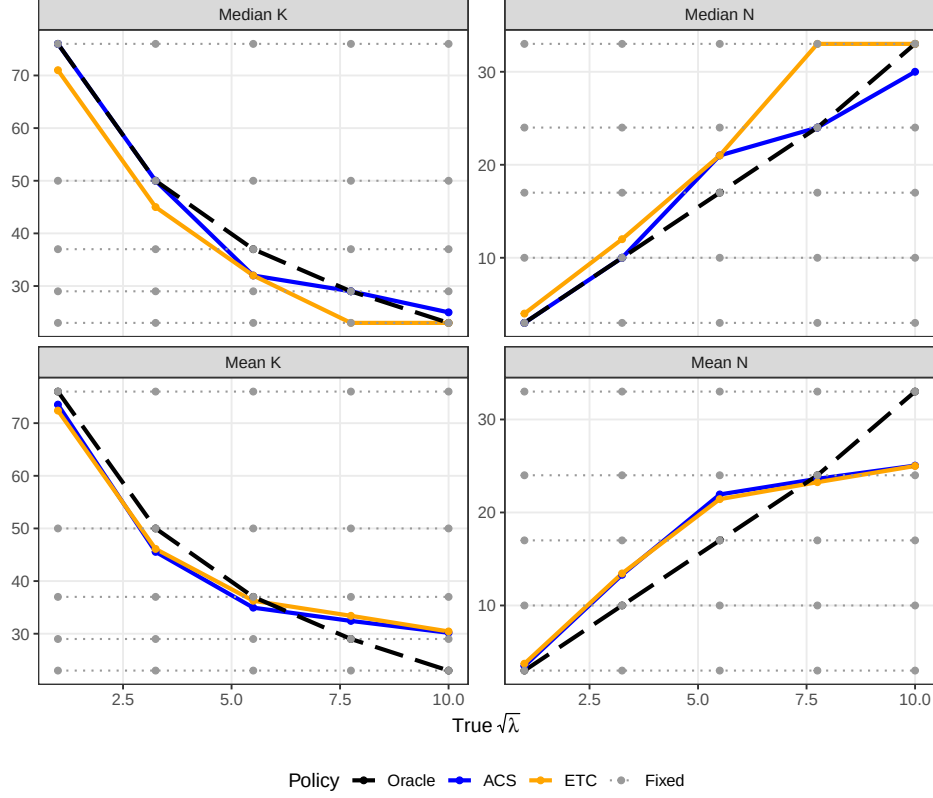


Figure 4: Median (top) and Mean (bottom) (K, N) across Policies and λ_0 . Fixed policies set at $\lambda \approx (1, 10, 30, 60, 100)$. $\mu = 0, \rho_0 = 1$. $B = 2000, F = 20, V = 2$.

Next, we evaluate the excess variance of our algorithms compared to the oracle design and an array of fixed (misspecified) policies. To do so, Figure 5 plots the mean and median excess variance of all policies in a λ grid. The red-dots in the pictures mark the worst median excess variance of each fixed policy, which naturally take place in the boundary. One can extract several takeaways from this graph. Firstly, both EtC-CS and ACS perform very well compared to the oracle and the alternative fixed policies. Secondly, this improved performance is particularly salient when assessing the median excess variance, what is to be expected as our proposed algorithms are random by nature, therefore, they occasionally incur large excess variance compared to fixed designs. Thirdly, our algorithms behave particularly well for small values of λ (large values of ρ), for the stability considerations discussed above. And, finally, it comes clear that our algorithms are the most robust to variations in λ . While fixed policies may outperform EtC-CS and ACS in small samples, their worst mean and median excess variance always improves on the worst mean and median excess variance of fixed competitors.

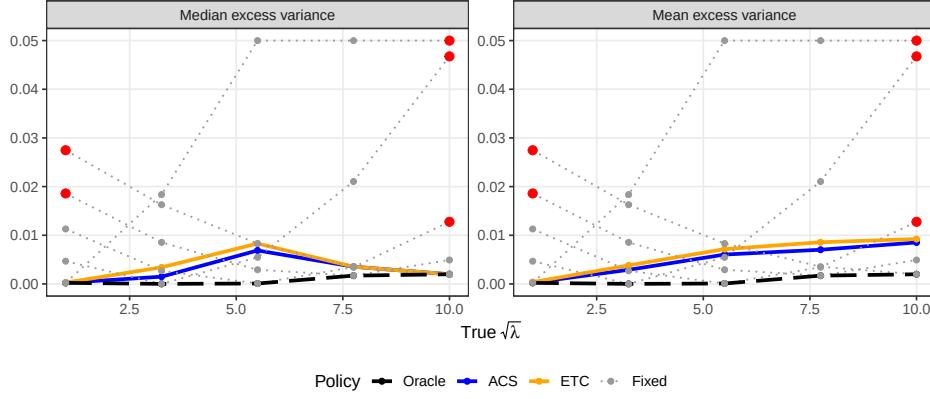


Figure 5: Median (left) and Mean (right) Excess Variance across Policies and λ_0 . Fixed policies set at $\lambda \approx (1, 10, 30, 60, 100)$. $\mu = 0, \rho_0 = 1$. $B = 2000, F = 20, V = 2$.

Finally, Figure 6 plots the evolution of excess variance as a function of budget, the main theoretical prescription in Section 4. For this particular parameter combination, (i) both **EtC-CS** and **ACS** display very low excess variance compared to the oracle starting at relatively modest budgets, (ii) both **EtC-CS** and **ACS** converge faster to zero excess variance than fixed policies (see Lemma 7.2), and (iii) **ACS** outperforms **EtC-CS** once again.

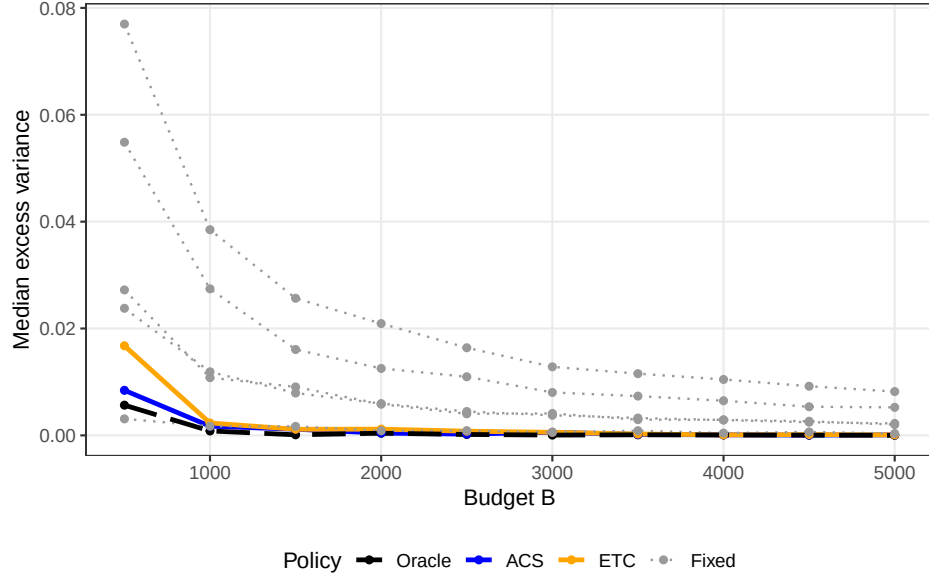


Figure 6: Median Excess Variance across Policies and Budget B . Fixed policies set at $\lambda \approx (1, 10, 30, 60, 100)$. $\mu = 0, \omega_0 = (1, 5)$. $F = 20, V = 2$.

To conclude this Section, we provide simulation evidence on conditional inference for **ACS**. Similar evidence for **EtC-CS** can be found in the Supplementary Material.²³ For a wide array of simulations satisfying symmetry, including bi-modal distributions and heavy-tailed (Laplace) distributions, I run a large number of simulations and selected the modal design for each of them to maximize the

²³Establishing conditional independence was far more challenging for **ACS** than it was for **EtC-CS**, as one had to show independence of $\hat{\mu}(K, N)$ and the entire σ -algebra $\sigma(\hat{\lambda}_1, \dots, \hat{\lambda}_\tau)$, as opposed to one-shot independence. As such, we consider this simulation evidence to be more suggestive of the theory results in Theorem 6.2.

number of observations at display. Figure 7 plots the empirical density for each distribution in the grid alongside the gaussian pdf. The first distribution in the title makes reference to the distribution of u_k and the second to the distribution of ε_{ki} . Similarly, for two distributions in the grid (the normal-normal and the bimodal-laplace), for each K , Figure 8 computes the share of observations of in $\pm 1.96\sqrt{\hat{b}(K^\pi, N^\pi)/K^\pi}$.

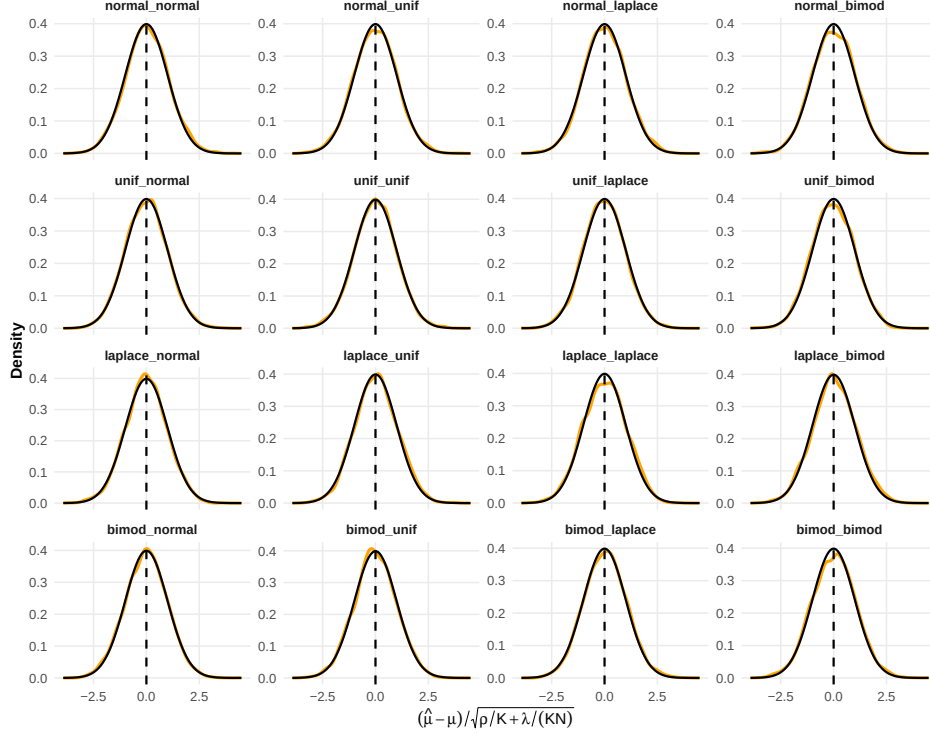


Figure 7: Empirical pdf for Modal Design Conditional Sample Average $(\hat{\mu}(K^\pi, N^\pi) - \mu) / \sqrt{\rho/K + \lambda/(K^\pi N^\pi)}$ across Symmetric Distributions. Left stands for u_k distribution. Right stands for ε_{ki} distribution. 10,000 simulations. $\mu = 0$, $\omega_0 = (1, 5)$. $B = 2000$, $F = 20$, $V = 2$.

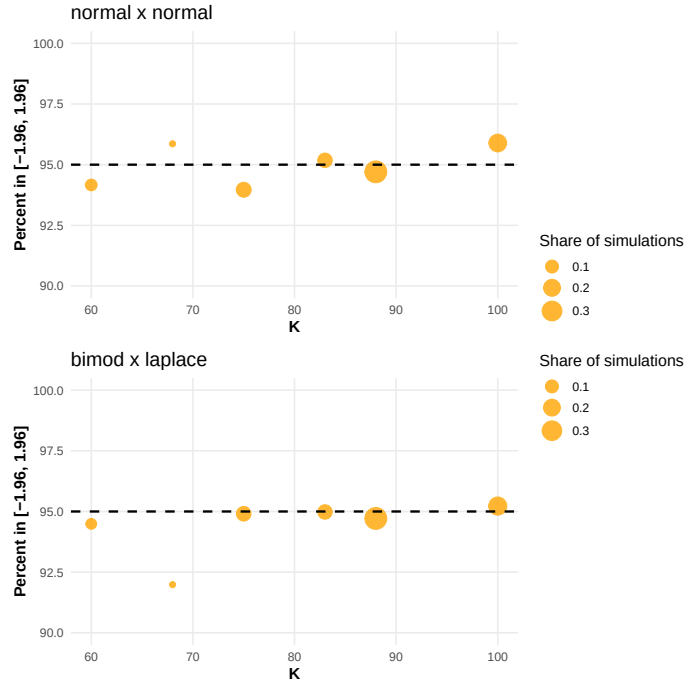


Figure 8: 95% Coverage for Design Conditional Sample Averages ($\hat{\mu}(K^\pi, N^\pi) - \mu)/\sqrt{\rho/K^\pi + \lambda/(K^\pi N^\pi)}$ with Normal-Normal (top) and Bimodal-Laplace (bottom) errors. 10,000 simulations. Only designs with 200+ observations are plotted. $\mu = 0$, $\omega_0 = (1, 5)$. $B = 2000$, $F = 20$, $V = 2$.

Both graphs provide compelling evidence of design-conditional independence in alignment with the results in Theorem 6.2. Table 1 reports the statistic and p-values of an Anderson-Darling (AD) normality test. As we can see, at standard confidence values, we cannot reject normality on any of them.²⁴

Error family	N obs	AD stat	p-value		Error family	n	AD stat.	p-value
Lap $\times$ Lap	3374	0.683	0.074		Normal $\times$ Normal	3554	0.296	0.594
Normal $\times$ Lap	3612	0.529	0.177		Unif $\times$ Normal	3677	0.290	0.612
Unif $\times$ Bimod	3615	0.419	0.327		Bimod $\times$ Unif	3566	0.268	0.683
Unif $\times$ Unif	3642	0.377	0.410		Bimod $\times$ Normal	3615	0.264	0.699
Normal $\times$ Bimod	3552	0.337	0.505		Bimod $\times$ Lap	3725	0.260	0.713
Bimod $\times$ Bimod	3527	0.302	0.576		Lap $\times$ Bimod	3305	0.246	0.757
Lap $\times$ Normal	3336	0.299	0.584		Normal $\times$ Unif	3561	0.238	0.783
Lap $\times$ Unif	3287	0.299	0.585		Unif $\times$ Lap	3675	0.199	0.885

Table 1: Anderson–Darling Normality Tests for the Modal Conditional Distribution of Standardized $\hat{\mu}(K^\pi, N^\pi)$ by Error Family (as in Figure 7). Left stands for u_k distribution. Right stands for ε_{ki} distribution. 10,000 Simulations. $\mu = 0$, $\omega_0 = (1, 5)$, $B = 2000$, $F = 20$, $V = 2$.

Finally, Figure 9 replicates the empirical pdf for a set of distributions that do not satisfy symmetry for at least one of the errors. We can reject the normality null for 2 out 3 distributions (p-values equal 0.0182, 0.0400 and 0.670, respectively). However, this is insufficient evidence to argue that

²⁴The AD test is relatively *overpowered* for large samples, as minimal tail deviations can lead to rejection. Moreover, the reported p-values do not apply any Bonferroni correction, despite the multiple hypothesis testing. Overall, we find this to be compelling evidence of design conditional normality.

symmetry is necessary for design conditional independence. Overall, we expect skewed-distributions to display slower convergence rates to normality (under a Berry-Esseen argument). When increasing the budget from 2,000 up to 5,000, we fail to reject normality for all three distributions at standard confidence levels, although p-values remain relatively low compared to the ones in Table 1. Especially, for the exponential-exponential family with a p-value ≈ 0.08 . In light of the simulation evidence, one may conjecture that symmetry is not a necessary condition for conditional asymptotic normality, yet it seems to play an important role in converging rates. Further study is granted.

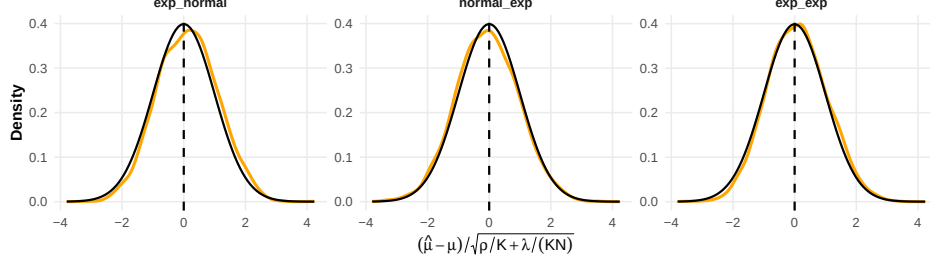


Figure 9: Empirical pdf for Modal Design Conditional Sample Average $(\hat{\mu}(K^\pi, N^\pi) - \mu) / \sqrt{\rho/K^\pi + \lambda/(K^\pi N^\pi)}$ across Asymmetric Distributions. Left stands for u_k distribution. Right stands for ε_{ki} distribution. 10,000 simulations. $\mu = 0$, $\omega_0 = (1, 5)$. $B = 2000$, $F = 20$, $V = 2$.

9 Conclusion

This paper presents a first solution to the problem of clustered experimental and survey design under unknown intra-cluster correlation. In particular, it presents two algorithms with varying degrees of adaptability which turn-out to be minmax optimal in a setting with additive errors, linear costs and a fixed budget. The algorithms do not only display negligible excess variance compared to the oracle, but they also outperform significantly every misspecified static policy. In addition, under a symmetry assumption, we show that inference on sample averages is invariant to adaptive data collection depending on sample variances. This strong result confirms that the variance reduction of our algorithms do not compromise inference. Our estimators remain asymptotically normal conditional on the design. Not without limitations, this is an encouraging result for adaptive data collection, and an attempt to push further adaptive experimentation and adaptive data collection.

References

- Adusumilli, K. (2026). How to sample and when to stop sampling: The generalised wald problem and minimax policies. *Review of Economic Studies*, 93(1):1–34.
- Aldous, D. J. and Eagleson, G. K. (1978). On mixing and stability of limit theorems. *The Annals of Probability*, pages 325–331.
- Ashraf, N., Bandiera, O., Davenport, E., and Lee, S. S. (2020). Losing prosociality in the quest for talent? sorting, selection, and productivity in the delivery of public services. *American Economic Review*, 110(5):1355–1394.
- Bai, Y. (2022). Optimality of matched-pair designs in randomized controlled trials. *American Economic Review*, 112(12):3911–3940.
- Bai, Y., Jiang, L., Romano, J. P., Shaikh, A. M., and Zhang, Y. (2024). Covariate adjustment in experiments with matched pairs. *Journal of Econometrics*, 241(1):105740.
- Ballin, M. and Barcaroli, G. (2013). Joint determination of optimal stratification and sample allocation using genetic algorithm. *Survey Methodology*, 39(2):369–393.
- Barcaroli, G., Fasulo, A., Guandalini, A., and Terribili, M. D. (2022). Two-stage sampling design and sample selection with the r package r2beat. *arXiv preprint arXiv:2209.06532*.
- Bethel, J. (1989). Sample allocation in multivariate surveys. *Survey methodology*, 15(1):47–57.
- Biemer, P. P. and Lyberg, L. E. (2003). *Introduction to survey quality*. John Wiley & Sons.
- Campbell, M. K., Grimshaw, J. M., and Elbourne, D. R. (2004). Intraclass correlation coefficients in cluster randomized trials: empirical insights into how should they be reported. *BMC Medical Research Methodology*, 4(1):9.
- Canay, I. A., Romano, J. P., and Shaikh, A. M. (2017). Randomization tests under an approximate symmetry assumption. *Econometrica*, 85(3):1013–1030.
- Cesa-Bianchi, N., Colomboni, R., and Kasy, M. (2025). Adaptive maximization of social welfare. *Econometrica*, 93(3):1073–1104.
- Chow, S.-C. and Corey, R. (2011). Benefits, challenges and obstacles of adaptive clinical trial designs. *Orphanet journal of rare diseases*, 6(1):79.
- Cicchitelli, G., Herzel, A., and Montanari, G. E. (1992). *Il campionamento statistico*. il Mulino.
- Cytrynbaum, M. (2021). Optimal stratification of survey experiments. *arXiv preprint arXiv:2111.08157*.
- Cytrynbaum, M. (2024). Covariate adjustment in stratified experiments. *Quantitative Economics*, 15(4):971–998.
- Datta, D. K. and Rajagopalan, N. (1998). Industry structure and ceo characteristics: An empirical study of succession events. *Strategic Management Journal*, 19(9):833–852.

- Duflo, E. and Banerjee, A. (2017). *Handbook of field experiments*, volume 1. Elsevier.
- Hadad, V., Hirshberg, D. A., Zhan, R., Wager, S., and Athey, S. (2021). Confidence intervals for policy evaluation in adaptive experiments. *Proceedings of the national academy of sciences*, 118(15):e2014602118.
- Hall, P. and Heyde, C. C. (2014). *Martingale limit theory and its application*. Academic press.
- Häusler, E. and Luschgy, H. (2015). *Stable convergence and stable limit theorems*, volume 74. Springer.
- Heß, S., Jaimovich, D., and Schündeln, M. (2021). Development projects and economic networks: Lessons from rural gambia. *The Review of Economic Studies*, 88(3):1347–1384.
- Imbens, G., Qin, C., and Wager, S. (2025). Admissibility of completely randomized trials: A large-deviation approach. *arXiv preprint arXiv:2506.05329*.
- Karing, A. (2024). Social signaling and childhood immunization: A field experiment in sierra leone. *The Quarterly Journal of Economics*, 139(4):2083–2133.
- Kasy, M. and Sautmann, A. (2021). Adaptive treatment assignment in experiments for policy choice. *Econometrica*, 89(1):113–132.
- Kiefer, J. and Wolfowitz, J. (1960). The equivalence of two extremum problems. *Canadian Journal of Mathematics*, 12:363–366.
- Kish, L. (1965). Survey sampling. new york: John wesley & sons. *Am Polit Sci Rev*, 59(4):1025.
- Knox, S. A. and Chondros, P. (2004). Observed intra-cluster correlation coefficients in a cluster survey sample of patient encounters in general practice in australia. *BMC Medical Research Methodology*, 4(1):30.
- Lattimore, T. and Szepesvári, C. (2020). *Bandit algorithms*. Cambridge University Press.
- Li, D., Raymond, L., and Bergman, P. (2026). Hiring as exploration. *Review of Economic Studies*, 93(2):1200–1240.
- Lin, G. C. and Ho, S. P. (2003). China’s land resources and land-use change: insights from the 1996 land survey. *Land use policy*, 20(2):87–107.
- Mbiti, I., Muralidharan, K., Romero, M., Schipper, Y., Manda, C., and Rajani, R. (2019). Inputs, incentives, and complementarities in education: Experimental evidence from tanzania. *The Quarterly journal of economics*, 134(3):1627–1673.
- Meriggi, N. F., Voors, M., Levine, M., Ramakrishna, V., Kangbai, D. M., Rozelle, M., Tyler, E., Kallon, S., Nabieu, J., Cundy, S., et al. (2024). Last-mile delivery increases vaccine uptake in sierra leone. *Nature*, 627(8004):612–619.
- Muralidharan, K. and Niehaus, P. (2017). Experimentation at scale. *Journal of Economic Perspectives*, 31(4):103–124.
- Neyman, J. (1992). On the two different aspects of the representative method: the method of stratified

- sampling and the method of purposive selection. In *Breakthroughs in statistics: Methodology and distribution*, pages 123–150. Springer.
- Nie, X., Tian, X., Taylor, J., and Zou, J. (2018). Why adaptively collected data have negative bias and how to correct for it. In *International Conference on Artificial Intelligence and Statistics*, pages 1261–1269. PMLR.
- Rudelson, M. and Vershynin, R. (2013). Hanson-wright inequality and sub-gaussian concentration.
- Sedgwick, P. (2015). Multistage sampling. *Bmj*, 351.
- Tam, B. (2005). Household sample surveys in developing and transition countries.
- Tschprow, A. (1923). On the two different aspects of the representative method: the method of stratified son the mathematical expectation of the moments of frequency distributions in the case of correlated observation sampling and the method of purposive selection. *Metron*, 2(646-683):192.
- Wager, S. (2024). Causal inference: A statistical learning approach. URL https://web.stanford.edu/swager/causal_inf_book.pdf.

A Proofs Section 3

Generalized Problem. We first introduce the general problem where the learner can sample an arbitrary number of units N_k from each cluster k subject to budget constraints. Let $\mathbf{N}$ be the vector of non-empty samples, i.e. $\mathbf{N} = (N_1, \dots, N_k, \dots, N_K)$. In this setting, the estimator of interest is given by the average of averages

$$\hat{\mu}(K, \mathbf{N}) = \frac{1}{K} \sum_{k=1}^K \frac{1}{N_k} \sum_{i=1}^{N_k} y_{ki}$$

where $K = \sum_k \mathbb{1}(N_k > 0)$. The variance of the estimator is given by

$$\begin{aligned} W(K, \mathbf{N}, \omega) &= \text{Var}(\hat{\mu}(K, \mathbf{N})) = \text{Var}\left(\frac{1}{K} \sum_{k=1}^K \frac{1}{N_k} \sum_{i=1}^{N_k} y_{ki}\right) = \text{Var}\left(\mu + \frac{1}{K} \sum_{k=1}^K u_k + \frac{1}{K} \sum_{k=1}^K \frac{1}{N_k} \sum_{i=1}^{N_k} \varepsilon_{ki}\right) \\ &= \frac{K}{K^2} \rho + \frac{1}{K^2} \sum_{k=1}^K \frac{\sigma}{N_k} = \frac{\rho}{K} \left(1 + \frac{1}{K} \sum_{k=1}^K \frac{\lambda}{N_k}\right) = \frac{\rho}{K} \left(1 + \frac{\lambda}{N_H(\mathbf{N})}\right) \end{aligned}$$

where $N_H = \left(\frac{1}{K} \sum_{k=1}^K \frac{1}{N_k}\right)^{-1}$ is the harmonic mean. For $N_k = N$ the expression simplifies to Equation 2.

Variance Minimization. The generalized variance minimization problem in Equation 2 can be written as

$$\min_{K \geq 2, \mathbf{N}: N_k \leq K \geq 2} W(K, \mathbf{N}, \omega) \quad \text{s.t.} \quad KF + V \sum_{k=1}^K N_k \leq B$$

Optimal Map. Fix K and define $S = \sum_{k=1}^K N_k$. Minimizing the variance in terms of N_k reduces to minimize $\sum_{k=1}^K \frac{1}{N_k}$ subject to $S = \sum_{k=1}^K N_k$. The map $x \mapsto 1/x$ is strictly convex in $\mathbb{R}_+$, therefore (by Jensen's)

$$\frac{1}{K} \sum_{k=1}^K \frac{1}{N_k} \geq \frac{1}{\frac{1}{K} \sum_{k=1}^K \frac{1}{N_k}} = \frac{K}{S}$$

with equality iff $N_k = N = \frac{S}{K}$ for all $k \leq K$. So, for any fixed K , the variance is minimized by a balanced allocation $N_k = N$. Next, let $K^*(N) = \frac{B}{F+VN}$. So the variance simplifies to

$$W(K^*(N), N, \omega) = \frac{(F + VN)\rho}{B} \left(1 + \frac{\lambda}{N}\right)$$

First order condition returns

$$\begin{aligned}
\frac{V\rho}{B} \left(1 + \frac{\lambda}{N}\right) &= \frac{(F + VN)\rho}{B} \left(\frac{\lambda}{N^2}\right) \\
V(N + \lambda) &= \frac{(F + VN)\lambda}{N} \\
N^*(\omega) = N^*(\lambda) &= \sqrt{\lambda \frac{F}{V}}
\end{aligned}$$

We further observe that $W(K^*(N), N, \omega)$ is strictly convex in N for $N \geq 0$, i.e.

$$W_{NN}(K^*(N), N, \omega) = -\frac{V\rho}{B} \frac{\lambda}{N^2} + \frac{V\rho\lambda}{BN^2} + \frac{(F + VN)\rho}{B} \frac{\lambda}{N^3} = \frac{(F + VN)\rho}{B} \frac{\lambda}{N^3} > 0$$

Thus the local point satisfying the FOC is indeed the global minimum of the function. Finally, because $N^*(\lambda)$ is strictly monotonic in λ with $\lim_{\lambda \rightarrow 0} N^*(\lambda) = 0$ and $\lim_{\lambda \rightarrow \infty} N^*(\lambda) = \infty$, the domain constraints on N lead to the final form of $N^*(\lambda)$. It follows that the optimal map (and oracle allocation) of the generalized problem is the same than the optimal map (and oracle allocation) of the simplified problem.

Oracle Variance. Assuming an interior solution for $N^*(\omega)$ (i.e. see Assumption 3.2),

$$K^*(\omega) = \frac{B}{F + \sqrt{FV\lambda}}$$

And,

$$\begin{aligned}
W(K^*(\lambda), N^*(\lambda), \omega) &= \frac{\rho}{K^*(\lambda)} \left(1 + \frac{\lambda}{N^*(\lambda)}\right) = \frac{(F + \sqrt{FV\lambda})\rho}{B} \left(1 + \sqrt{\lambda \frac{V}{F}}\right) \\
&= \frac{\rho}{B} (F + 2\sqrt{F\lambda V} + \lambda V) = \frac{\rho (\sqrt{F} + \sqrt{\lambda V})^2}{B}
\end{aligned}$$

B Proofs of Section 4

Proof of Lemma 4.1. Unbiasedness of the estimators. To be used for a LLN argument,

$$\begin{aligned}
\mathbb{E}[\tilde{w}(K, N)] &= \frac{1}{K(N-1)} \sum_{k=1}^K \sum_{i=1}^N \mathbb{E}[(y_{ki} - \hat{\mu}^k(N))^2] = \frac{1}{K(N-1)} \sum_{k=1}^K \sum_{i=1}^N \text{Var}(y_{ki} - \hat{\mu}^k(N)) \\
&= \frac{1}{K(N-1)} K(N-1)\sigma = \sigma \\
\mathbb{E}[\tilde{b}(K, N)] &= \frac{1}{K-1} \sum_{k=1}^K \mathbb{E}[(\hat{\mu}^k(N) - \hat{\mu}(K, N))^2] = \frac{1}{K-1} (K-1) \left(\rho + \frac{\sigma}{N}\right) = \rho + \frac{\sigma}{N} \\
\mathbb{E}[\tilde{\rho}(K, N)] &= \mathbb{E}[\hat{b}(K, N)] - \mathbb{E}\left[\frac{\hat{w}(K, N)}{N}\right] = \rho + \frac{\sigma}{N} - \frac{\sigma}{N} = \rho
\end{aligned}$$

Consistency,

$$\tilde{w}(K, N) = \frac{1}{K} \sum_{k=1}^K \frac{1}{N-1} \sum_{i=1}^N (y_{ki} - \hat{\mu}^k(N))^2 \xrightarrow{p} \sigma$$

where convergence follows from the LLN. We used cross-cluster independence to argue that that $\sum_{i=1}^N (y_{ki} - \hat{\mu}^k(N))^2 \perp \sum_{i=1}^N (y_{li} - \hat{\mu}^l(N))^2$ for $k \neq l$. A similar argument holds for consistency of $\tilde{w}(K, N)$ under N asymptotics, namely

$$\begin{aligned} \tilde{w}(K, N) &= \frac{1}{N-1} \sum_{i=1}^N \frac{1}{K} \sum_{k=1}^K (y_{ki} - \hat{\mu}^k(N))^2 = \frac{1}{N-1} \sum_{i=1}^N \frac{1}{K} \sum_{k=1}^K (\varepsilon_{ki} - \bar{\varepsilon}_{ki})^2 \\ &= \frac{N}{N-1} \left[\frac{1}{N} \sum_{i=1}^N \varepsilon_{ki}^2 - \bar{\varepsilon}_{ki}^2 \right] \xrightarrow{p} \sigma \end{aligned}$$

provided $\frac{1}{N} \sum_{i=1}^N \varepsilon_{ki}^2 \xrightarrow{p} \sigma$ under the LLN given ε_{ki} is iid and $\bar{\varepsilon}_{ki}^2 \xrightarrow{p} 0$ also under the LLN. Applying CMT completes the proof. Similarly,

$$\tilde{b}(K, N) = \frac{1}{K-1} \sum_{k=1}^K (\hat{\mu}^k(N) - \hat{\mu}(K, N))^2 \xrightarrow{p} \rho + \frac{\sigma}{N}$$

where we used the same decomposition and argue that $\bar{u}_k + \bar{\varepsilon}_{ki} \xrightarrow{p} 0$. Consistency of $\tilde{\rho}(K, N)$ holds by the CMT and the consistency of $\tilde{b}(K, N)$ and $\tilde{w}(K, N)$ (only under K asymptotics). In addition, note that projection $\Pi_{[a,b]}(x)$ onto a closed interval $[a, b]$ is 1-Lipschitz, therefore convergence of $|\tilde{w}(K, N) - \sigma|$ implies convergence of $|\hat{w}(K, N) - \sigma|$. The same logic applies for $\hat{\rho}(K, N)$. Finally, consistency of $\hat{\lambda}(K, N)$ follows from consistency of $\hat{w}(K, N)$ and $\hat{\rho}(K, N)$ under CMT. $\square$

C Proofs of Section 5.

This Section is structured as follows. We first present a set of preliminary lemmas that will play an important role both in the derivation of high probability upper bounds and our inference results. We then discuss Theorem 5.2 by first presenting a general lemma from which the theorem will follow as a simple corollary. Finally, we transform our high probability bounds into bounds in expectation by deriving worst-case upper bounds.

C.1 Preliminary Lemmas and Definitions

Lemma C.1. *Implementation of Optimal Designs. Let (K^π, N^π) be the design induced by policy π . Then, under Assumption 3.1 and 3.2, $(K^{\text{EtC-CS}}, N^{\text{EtC-CS}}) = (K^*(\hat{\lambda}(K_M, N_m)), N^*(\hat{\lambda}(K_M, N_m)))$ and $(K^{\text{ACS}}, N^{\text{ACS}}) = (K^*(\hat{\lambda}(\tau-1, N_m)), N^*(\hat{\lambda}(\tau-1, N_m)))$ a.s., where τ is defined as in Equation 4.*

Proof of Lemma C.1. In the case of EtC-CS, the Lemma is a simple restatement of the algorithm under the assumptions. In the case of ACS, by the estimator's clipping $N_p = N^*(\lambda_m) \leq N^*(\hat{\lambda}(K, N_p))$ for all $K \in [K_M, K_m]$. It follows that for some stopping time τ , $K^*(\hat{\lambda}(\tau-1, N_p)) = \frac{B}{F+VN^*(\hat{\lambda}(\tau-1, N_p))} \leq \frac{B}{F+VN_p}$. Therefore, up to integer considerations, $\tau = \inf\{k : k+1 > K^*(\hat{\lambda}(k, N_p))\}$. In other words, the algorithm stops before exhausting the budget. Moreover, at time $\tau-1$, $\tau \leq K^*(\hat{\lambda}(\tau-1, N_p))$, otherwise the algorithm would have not explored cluster τ . It follows that the design $(K^*(\hat{\lambda}(\tau-1, N_p)), N^*(\hat{\lambda}(\tau-1, N_p)))$ is budget feasible (provided $\tau \leq K^*(\hat{\lambda}(\tau-1, N_p))$ and

$$N_p \leq N^*(\hat{\lambda}(\tau - 1, N_p)).$$

■

Lemma C.2. *Second Order Approximation*

For any policy π which induces $(K^\pi, N^\pi) = (K^*(\hat{\lambda}(K, N)), N^*(\hat{\lambda}(K, N)))$ a.s. with $(K, N) \geq 2$, then

$$\Delta_\pi = \frac{1}{2} W_{\lambda\lambda}(\omega_0)(\hat{\lambda}(K, N) - \lambda_0)^2 + R_p$$

where $W_{\lambda\lambda}(\omega_0) \sim O(1/B)$ and $R_p \sim O((\hat{\lambda}(K, N) - \lambda_0)^3)$.

Proof of Lemma C.2. Lemma follows from a simple second order approx around λ_0

$$\begin{aligned} \Delta_\pi &= W(K^\pi, N^\pi, \omega_0) - W(K^*(\lambda_0), N^*(\lambda_0), \omega_0) \\ &= W(K^*(\hat{\lambda}(K, N)), N^*(\hat{\lambda}(K, N)), \omega_0) - W(K^*(\lambda_0), N^*(\lambda_0), \omega_0) \\ &= W_\lambda(\omega_0)(\hat{\lambda}(K, N) - \lambda_0) + \frac{1}{2} W_{\lambda\lambda}(\omega_0)(\hat{\lambda}(K, N) - \lambda_0)^2 + O((\hat{\lambda}(K, N) - \lambda_0)^3) \\ &= \frac{1}{2} W_{\lambda\lambda}(\omega_0)(\hat{\lambda}(K, N) - \lambda_0)^2 + O((\hat{\lambda}(K, N) - \lambda_0)^3) \end{aligned}$$

where the last equality follows from optimality of (K^*, N^*) at λ_0 and the Envelope Theorem. Finally, we show analytically that $W_{\lambda\lambda}(\omega_0) \sim O(1/B)$

$$\begin{aligned} W(K^*(\lambda), N^*(\lambda), \omega_0) &= \rho_0 \frac{F + V\sqrt{\lambda \frac{F}{V}}}{B} \cdot \left(1 + \frac{\lambda_0}{\sqrt{\lambda \frac{F}{V}}} \right) = \frac{\rho_0}{B} \cdot \left(F + \lambda_0 V + \sqrt{FV} \left(\lambda_0 / \lambda^{1/2} + \lambda^{1/2} \right) \right) \\ W_\lambda(K^*(\lambda), N^*(\lambda), \omega_0) &= \frac{\rho_0}{2B} \sqrt{FV} \left(-\lambda_0 \lambda^{-3/2} + \lambda^{-1/2} \right) \\ W_{\lambda\lambda}(K^*(\lambda), N^*(\lambda), \omega_0) &= \frac{\rho_0}{4B} \sqrt{FV} \left(3\lambda_0 \lambda^{-5/2} - \lambda^{-3/2} \right) \\ W_{\lambda\lambda}(K^*(\lambda_0), N^*(\lambda_0), \omega_0) &= \frac{\rho_0}{4B} \sqrt{FV} \left(3\lambda_0^{-3/2} - \lambda_0^{-3/2} \right) = \frac{\rho_0 \sqrt{FV}}{2B\lambda_0^{3/2}} \end{aligned}$$

■

Lemma C.3. *Consistent Exploration and Consistent Design.* Let (K_p^π, N_p^π) be the pilot design induced by policy π and let (K^π, N^π) be the design induced by policy π . Define $\kappa(\lambda) = K^*(\lambda)/B$, $\kappa_0 = \kappa(\lambda_0)$ and $N_0 = N^*(\lambda_0)$. Then, under Assumptions 3.1 and 3.2,

$$\begin{aligned} (K_p^{EtC-CS}/B, N_p^{EtC-CS}) &\xrightarrow{p} (\kappa_p, N_p) & (K^{EtC-CS}/B, N^{EtC-CS}) &\xrightarrow{p} (\kappa_0, N_0) \\ (K_p^{ACS}/B, N_p^{ACS}) &\xrightarrow{p} (\kappa_0, N_p) & (K^{EtC-CS}/B, N^{EtC-CS}) &\xrightarrow{p} (\kappa_0, N_0) \end{aligned}$$

Interpretation of Lemma C.3. The pilot section of the Lemma can be seen as a stronger

version of 4.2. The Lemma states that both EtC-CS and ACS implement consistent pilots under B asymptotics, and that they implement the optimal design $(K^*(\lambda_0), N^*(\lambda_0))$, also under B asymptotics. Pilot consistency for EtC-CS is trivial, provided the pilot is not random. Moreover, under Lemma C.1, EtC-CS implements an optimal map of the form $(K^*(\hat{\lambda}(K_M, N_m)), N^*(\hat{\lambda}(K_M, N_m)))$ where $K_M = B/(F + VN_M)$, therefore, under Lemma 4.1 $\hat{\lambda}(K_M, N_m) \rightarrow \lambda_0$. From the CMT, $(K^*(\hat{\lambda}(K_M, N_m))/B, N^*(\hat{\lambda}(K_M, N_m))) \rightarrow (\kappa_0, N_0)$.

The proof of the Lemma for ACS follows from noting that, by definition of the clipping estimator and the Assumptions in the Lemma, $\tau \in [K_M, K_m]$ almost surely. Then, by Lemmas C.1, 4.1, and the CMT, $(K_p^{\text{ACS}}/B, N_p^{\text{ACS}}) \rightarrow (\kappa_0, N_p)$ (with N_p fixed and independent of B). Similarly, $(K^{\text{ACS}}/B, N^{\text{ACS}}) \rightarrow (\kappa_0, N_0)$. Note, that ACS (unlike EtC-CS) samples all clusters within the pilot.

C.2 Definitions

For simplicity, write a_{K_p} for any estimator of the form $a(K, N)$ with N fixed. Define

$$\varepsilon_k = (\varepsilon_{1k}, \dots, \varepsilon_{Kk}) \quad \bar{\varepsilon}_k^N = \frac{1}{N} \sum_{i=1}^N \varepsilon_{ki} \quad W_k = \frac{1}{N_p - 1} \sum_{i=1}^{N_p} (\varepsilon_{ki} - \bar{\varepsilon}_k^{N_p})^2 \quad \tilde{w}_K = \frac{1}{K - 1} \sum_{k=1}^K W_k$$

where $\mathbb{E}[W_k] = \sigma$ and we define $A_k = W_k - \sigma$. In particular, note that W_k is the usual within-cluster k variance and $\tilde{w}_K$ is the average of within-cluster variances, which also identifies the within-cluster variance for every k (as in Equation 4.1). Now, observe that

$$\hat{\mu}(K, N) - \mu_0 = \frac{1}{K} \sum_{k=1}^K M_{k,N} \quad M_{k,N} = u_k + \bar{\varepsilon}_k^N \quad \bar{M}_{K,N} = \frac{1}{K} \sum_{k=1}^K M_{k,N}$$

where $M_{k,N}$ is the average of cluster k errors such that $\mathbb{E}[M_{k,N}] = 0$. Similarly, just like we did with the within-cluster variance $\hat{w}_k$,

$$\tilde{b}_K = \frac{1}{K - 1} \sum_{k=1}^K (M_{k,N_p} - \bar{M}_{K,N_p})^2$$

where $\tilde{b}_K$ is the between-cluster variance (as in Equation 4.1) such that $\mathbb{E}[\tilde{b}_K] = \rho + \sigma/N_p = \theta_{N_p}$. In addition, define the centered second moment of the average of cluster k errors

$$B_k = M_{k,N_p}^2 - \theta_{N_p}$$

Unlike $\tilde{w}_k$, $\tilde{b}_k$ cannot be written as the average of B_k as $\tilde{b}_k$ demeans M_{k,N_p} by the (residuals of the) grand mean $\bar{M}_{K,N_p}$, yet we can decompose $\tilde{b}_K - \theta_{N_p}$ in the following way,

$$\begin{aligned}
\tilde{b}_K - \theta_{N_p} &= \frac{1}{K-1} \sum_{k=1}^K (M_{k,N_p} - \bar{M}_{K,N_p})^2 - \theta_{N_p} \\
&= \frac{1}{K-1} \left(\sum_{k=1}^K M_{k,N_p}^2 - K \bar{M}_{K,N_p}^2 \right) - \theta_{N_p} \\
&= \frac{1}{K-1} \left(K \theta_{N_p} + \sum_{k=1}^K (M_{k,N_p}^2 - \theta_{N_p}) - K \bar{M}_{K,N_p}^2 \right) - \theta_{N_p} \\
&= \frac{1}{K-1} \sum_{k=1}^K B_k + \frac{\theta_{N_p}}{K-1} - \frac{K}{K-1} \bar{M}_{K,N_p}^2
\end{aligned}$$

C.3 Concentration Inequalities

Subgaussian and Subexponential Constants. We now discuss the concentration properties of M_k, A_k, B_k in light of Assumption 5.1. By independence of $(u_k, \bar{\varepsilon}_k)$, we can immediately recover that M_k is $\sqrt{\kappa_u + \kappa_\varepsilon/N_p}$ -subgaussian. Therefore, it is subexponential with parameters $\nu_M = \kappa_u + \kappa_\varepsilon/N_p, b_M \rightarrow 0$. B_k is the square of a subgaussian random variable, therefore it is subexponential with parameters $\nu_B = (\kappa_u + \kappa_\varepsilon/N_p)^2, b_B = \kappa_u + \kappa_\varepsilon/N_p$. Finally, note that

$$A_k = \frac{1}{N_p - 1} (\varepsilon'_k \mathbf{B}_{N_p} \varepsilon_k - (N_p - 1)\sigma)$$

where $\mathbf{B}_{N_p} = I_{N_p} - \frac{1}{N_p} \mathbf{1}\mathbf{1}'$. Then, by Hanson-Wright (see Rudelson and Vershynin (2013)), A_k is subexponential with parameters $\nu_A = \kappa_\varepsilon / \|\mathbf{B}\|_F^2$ and $b_A = \kappa_\varepsilon / \|\mathbf{B}\|$, where $\|\cdot\|_F$ is the Frobenius norm and $\|\cdot\|$ is the operator norm. Because $\mathbf{B}$ is an orthogonal projection of rank $N_p - 1$, $\|\mathbf{B}\|_F^2 = (N_p - 1)$ and $\|\mathbf{B}\| = 1$. It follows that A_k is subexponential with parameters $\nu_A = \kappa_\varepsilon / (N_p - 1)$ and $b_A = \kappa_\varepsilon$.

Maximal Bernstein. We now present without proof Bernstein's maximal inequality for sums of *iid* subexponential variables. This maximal inequality will play a central role in the derivation of Lemma C.4. Let $\{\xi_k\}$ be a sequence of *iid* (ν, b) -subexponential random variables. Then, for every $\bar{K} \geq 1$ and $t > 0$,

$$\mathbb{P} \left(\max_{1 \leq K \leq \bar{K}} \sum_{k=1}^K \xi_k \geq t \right) \leq 2 \exp \left(-\frac{1}{2} \min \left(\frac{t^2}{\bar{K} \nu^2}, \frac{t}{b} \right) \right)$$

C.4 High Probability Upper Bounds

Lemma C.4. *Generalized Theorem 5.2. Assume that a policy π induces a pilot of random size (K_p^π, N_p^π) with $K_p^\pi \in \mathcal{K} = [\underline{K}, \bar{K}]$ a.s. with $\bar{K}/\underline{K} \leq C_0 = O(1) \perp B$ and $N_p^\pi = N_p$ fixed. Under Assumptions 3.1, 3.2 and 5.1, there exists a positive finite constant C not depending on B such that*

$$\mathbb{P}(|\hat{\lambda}_{K_p^\pi} - \lambda_0| \leq D^\lambda(\alpha) \ \forall K \in \mathcal{K}) \geq 1 - \alpha, \quad D^\lambda(\alpha) = C \sqrt{\frac{\ln(1/\alpha)}{\underline{K}}}$$

Proof of Lemma C.4. For simplicity, for the remainder of the proof, we leave aside distinctions between K and $K - 1$ in the definition of the estimators provided these differences will only contribute an order-negligible excess variance to the final results and the uncorrected estimators remain consistent as in Lemma 4.1. We now take A_k as an example (with subexponential constants (ν_A, b_A)). The same argument will go through for B_k and M_k . Define $\delta(\nu, \alpha) = \nu \sqrt{\frac{2C_0 \ln(6/\alpha)}{\underline{K}}}$

Start by considering Bernstein's Maximal Inequality with $t = \delta \underline{K}$, then

$$\begin{aligned} \mathbb{P}(|\tilde{w}_{K_p^\pi} - \sigma| \geq \delta(\nu_A, b_A)) &\leq \mathbb{P}\left(\sup_{\underline{K} \leq K \leq \overline{K}} |\tilde{w}_K - \sigma| \geq \delta(\nu_A, b_A)\right) \\ &\leq \mathbb{P}\left(\frac{1}{\underline{K}} \max_{\underline{K} \leq K \leq \overline{K}} \left|\sum_{k=1}^K A_k\right| \geq \delta(\nu_A, b_A)\right) \\ &\leq 2 \exp\left(-\frac{1}{2} \min\left(\frac{\delta(\nu_A, b_A)^2 \underline{K}^2}{\overline{K} \nu_A^2}, \frac{t}{b_A}\right)\right) \\ &\leq 2 \exp\left(-\frac{1}{2} \min\left(\frac{\delta(\nu_A, b_A)^2 \underline{K}}{C_0 \nu_A^2}, \frac{t}{b_A}\right)\right) \leq \alpha/3 \end{aligned}$$

For consistency with the notation in the online learning literature, define $\delta(\nu_A, b_A) = D^w(\alpha) \sim O\left(\sqrt{\frac{\ln(1/\alpha)}{\underline{K}}}\right)$. A similar argument shows that $\mathbb{P}\left(\frac{1}{K_p^\pi} \sum_{k=1}^{K_p^\pi} M_k \geq \delta(\nu_M, b_M)\right) \leq \alpha/3$ and $\mathbb{P}\left(\frac{1}{K_p^\pi} \sum_{k=1}^{K_p^\pi} B_k \geq \delta(\nu_B, b_B)\right) \leq \alpha/3$. Define the good event $\mathcal{E}$ as the intersection of the negation of the 3 events above, then $\mathbb{P}(\mathcal{E}) \geq 1 - \alpha$. Moreover, under event $\mathcal{E}$,

$$\begin{aligned} |\tilde{b}_{K_p^\pi} - \theta| &= \left| \frac{1}{K_p^\pi} \sum_{k=1}^{K_p^\pi} B_k + \frac{\theta}{K_p^\pi} - \left(\frac{1}{K_p^\pi} \sum_{k=1}^{K_p^\pi} M_k \right) \right|^2 \\ &\leq \delta(\nu_B) + \delta^2(\nu_M) + \left| \frac{\theta}{\underline{K}} \right| \leq D^b(\alpha) \end{aligned}$$

where $D^b(\alpha) \sim O\left(\sqrt{\frac{\ln(1/\alpha)}{\underline{K}}}\right)$ (provided $\frac{\theta}{K_p} \sim O(1/\underline{K})$). We can now apply the triangle inequality to obtain a similar bound for $|\tilde{\rho} - \rho|$ under $\mathcal{E}$, i.e.

$$\begin{aligned} |\tilde{\rho}_{K_p^\pi} - \rho| &= |\tilde{b}_{K_p^\pi} - \tilde{w}_{K_p^\pi}/N_p - \rho| = |\tilde{b}_{K_p^\pi} - (\rho + \sigma/N_p) - 1/N_p(\tilde{w}_{K_p^\pi} - \sigma)| \\ &\leq |\tilde{b}_{K_p^\pi} - (\rho + \sigma/N_p)| + \frac{1}{N_p} |\tilde{w}_{K_p^\pi} - \sigma| \leq D^b(\alpha) + \frac{1}{N_p} D^w(\alpha) = D^\rho(\alpha) \end{aligned}$$

with $D_\alpha^\rho \sim O\left(\sqrt{\frac{\ln(1/\alpha)}{\underline{K}}}\right)$. Again, clipping preserves the same bounds deterministically for $\hat{\rho}_{K_p^\pi}$. A similar trick allows us to upper bound deviations for $\hat{\lambda}_{K_p^\pi}$ uniformly over $[\underline{K}, \overline{K}]$ under $\mathcal{E}$, i.e.

$$\begin{aligned}
|\hat{\lambda}_{K_p^\pi} - \lambda| &= \left| \frac{\hat{w}_{K_p^\pi} - \sigma}{\hat{\rho}_{K_p^\pi}} \right| = \left| \frac{\rho \hat{w}_{K_p^\pi} - \sigma \hat{\rho}_{K_p^\pi}}{\rho \hat{\rho}_{K_p^\pi}} \right| = \left| \frac{\rho (\hat{w}_{K_p^\pi} - \sigma) - \sigma (\hat{\rho}_{K_p^\pi} - \rho)}{\rho \hat{\rho}_{K_p^\pi}} \right| \\
&\leq \left| \frac{\rho (\hat{w}_{K_p^\pi} - \sigma) - \sigma (\hat{\rho}_{K_p^\pi} - \rho)}{\rho \rho_m} \right| \leq \frac{1}{\rho_m} (D^w(\alpha) + \lambda_M D^\rho(\alpha)) = D^\lambda(\alpha) \sim O\left(\sqrt{\frac{\ln(1/\alpha)}{K}}\right)
\end{aligned}$$

Completing the proof. ■

Proof of Theorem 5.2. Observe that both EtC-CS and ACS satisfy the assumptions in Lemma C.4. In particular, for EtC-CS, $K_p^\pi \in [K_M, K_M]$, a.s. with $C_0 = K_M/K_M = 1$, and for ACS $K_p^\pi = \tau \in [K_M, K_m]$ a.s. with $C_0 = K_m/K_M = O(1)$ (independent of B). See the remark in Lemma C.3. It then follows that under $\mathcal{E}$,

$$\begin{aligned}
\Delta_\pi &= \frac{C_1}{B} (\hat{\lambda}_{K_p^\pi} - \lambda_0)^2 + R_p \leq \frac{C_1}{B} D^\lambda(\alpha)^2 + R_p \\
&\leq \frac{C_2}{B} \frac{\ln(1/\alpha)}{K_m} + R_p \leq C_3 \frac{\ln(1/\alpha)}{B^2}
\end{aligned}$$

where the first equality follows from Lemma C.1, the second and third inequalities follow from Lemma C.4 under the good event $\mathcal{E}$, and the fourth inequality from noting that $K_m = \Theta(B) = \frac{B}{F+VN^*(\lambda_m)}$. Moreover, we used the fact that $R_p \sim O(B^{-5/2})$, thus this term gets absorbed by the constant C_3 . We note that both C_2 and C_3 depend on N_p and C_0 , but not on B . Theorem follows from noting that event $\mathcal{E}$ has probability greater or equal than $1 - \alpha$ as per Lemma C.4. □

Interpretation of Theorem 5.2. In a nutshell, Theorem says that for small enough pilots, both algorithms implement optimal maps for some $\hat{\lambda}(K_p^\pi, N_p^\pi)$. By letting K_p^π grow with budget, the learner can implement the oracle design in the limit. The proof is somewhat of an overkill for both EtC-CS and ACS but for different reasons. In the case of EtC-CS one needs to control deviations of $\hat{\lambda}$ for a deterministic pilot size, hence, a simple bernstein inequality would have sufficed (without invoking maximal bernstein nor the martingale properties of cumulative errors). This proof structure was in fact selected for ACS provided the pilot size is random under this policy. Yet, to effectively lower bound τ by K_M is deliberately lazy, as $\tau \rightarrow K^*(\lambda_0)$ (possibly much bigger than K_M) under Lemma C.3. Still, these bounds are essentially unimprovable as there is no order loss when considering maximal deviations in a window of length $O(1)$ (in the asymptotic regime).

C.5 Worst Case and Expected Excess Variance

Proof of Lemma 5.3. For all policies satisfying the budget constraint, the variance minimization problem admits a representation as a single variable unconstrained minimization exercise of the form $\min_{2 \leq N \leq (B-2F)/(2V)} W(K^*(N), N, \omega)$. As discussed in Appendix A, $W(K^*(N), N, \omega)$ is convex in N , therefore the maximum is achieved at the boundaries, i.e. $W(K^*(2), 2, \omega)$ or $W(2, (B-2F)/(2V), \omega)$. Our results would not change materially if the designer could select $N = 1$ or $N = (B-F)/V$ (i.e. $K = 1$). However, for consistency with the rest of the paper we keep the same domain. Now, simple calculations show that

$$\begin{aligned}
W(K^*(2), 2, \omega_0) - W(K^*(\lambda_0), N^*(\lambda_0), \omega_0) &= \frac{\rho_0}{2B} \left(\sqrt{\lambda_0 F} - 2\sqrt{V} \right)^2 \\
W(2, (B - 2F)/(2V), \omega_0) - W(K^*(\lambda_0), N^*(\lambda_0), \omega_0) &= \frac{\rho_0}{2} \left(1 + \frac{2\lambda_0 V}{B - 2F} \right) - \frac{\rho_0}{B} \left(\sqrt{F} + \sqrt{\lambda_0 V} \right)^2
\end{aligned}$$

While the first term converges to 0 as $B \rightarrow \infty$, the second term converges to $\frac{\rho_0}{2}$ from above. Therefore, the worst case excess variance is asymptotically upper bounded by $\frac{\rho_0}{2} + \delta$ for some small positive δ . $\square$

Proof of Theorem 5.4. Let $\alpha = \frac{C_\alpha}{B^2}$ for some finite C_α , then for any policy $\pi \in \{\text{EtC-CS, ACS}\}$,

$$\begin{aligned}
\mathbb{E}_\pi [\Delta_\pi] &= \mathbb{P}_\pi(\bar{\mathcal{E}}) \cdot \mathbb{E}_\pi [\Delta_\pi \mid \bar{\mathcal{E}}] + \mathbb{P}_\pi(\mathcal{E}) \cdot \mathbb{E}_\pi [\Delta_\pi \mid \mathcal{E}] \\
&\leq \alpha \left(\frac{\rho_M}{2} + \delta \right) + C_1 \frac{\ln(1/\alpha)}{B^2} \\
&= \frac{C_\alpha}{B^2} + C_1 \frac{\ln(B^2/C_\alpha)}{B^2} \leq C_2 \frac{\ln(B)}{B^2}
\end{aligned}$$

where the key inequality follows from Theorem 5.2 and Lemma 5.3. α has been selected such that the second expression is minimized. $\square$

D Proofs of Section 6

This section is structured by first presenting a set of preliminary lemmas that will be invoked in the proof Lemma D.5. Theorem 6.2 will follow as a corollary of the main lemma in this section.

D.1 Preliminary Lemmas

Lemma D.1. *Conditional Zero Mean.* Define $L_k = \frac{\theta_{N_P}}{\rho^2} A_k - \frac{\sigma}{\rho^2} B_k$

$$\mathbb{E}[M_{k,N} \mid L_k] = 0 \quad a.s$$

Proof of Lemma D.1. Consider the cluster error sign map $f(\{C_k\}) \mapsto f(\{-C_k\})$ for all k . Then, $M_{k,N} \mapsto -M_{k,N}$ (so $M_{k,N}$ is odd), while L_k provided $W_k \mapsto W_k$ and $B_k \mapsto B_k$, so they are even. Note that B_k is even as it is the square of an odd function minus a constant. Then, for any bounded ϕ

$$\mathbb{E}[M_{k,N} \phi(L_k)] = \mathbb{E}[-(M_{k,N}) \phi(L_k)] = -\mathbb{E}[M_{k,N} \phi(L_k)] \implies \mathbb{E}[M_{k,N} \phi(L_k)] = 0$$

where the first equality follows from Assumption 6.1.i. $\blacksquare$

Lemma D.2. *Approximate Form of $\hat{b}_K$.*

$$\sqrt{K}(\hat{b}_K - \theta_{N_p}) = \frac{1}{\sqrt{K}} \sum_{k=1}^K B_k + o_p(1)$$

Proof of Lemma D.2. Starting from the decomposition of $\hat{b}_K - \theta_{N_p}$ in the Upper Bound section,

$$\hat{b}_K - \theta_{N_p} = \frac{1}{K-1} \sum_{k=1}^K M_{k,N_p}^2 + \frac{\theta_{N_p}}{K-1} - \frac{K}{K-1} \bar{M}_{K,N_p}^2$$

It then follows that

$$\frac{\sqrt{K}}{K-1} \theta_{N_p} = O(K^{-1/2}) = o(1) \quad \frac{\sqrt{K}}{K-1} \sum_{k=1}^K M_{k,N_p}^2 = \frac{1}{\sqrt{K}} \sum_{k=1}^K M_{k,N_p}^2 + o_p(1) \quad \frac{K\sqrt{K}}{K-1} \bar{M}_{K,N_p}^2 = o_p(1)$$

where the second and third equality follow from the bounded moments assumption in Assumption 6.1.ii and noting that $\bar{M}_{K,N_p} = O_p(K^{-1/2})$, therefore $\bar{M}_{K,N_p}^2 = O_p(K^{-1})$, $\frac{K\sqrt{K}}{K-1} \bar{M}_{K,N_p}^2 = O_p(K^{-1/2}) = o_p(1)$. It follows that

$$\sqrt{K}(\hat{b}_K - \theta_{N_p}) = \frac{1}{\sqrt{K}} \sum_{k=1}^K B_k + o_p(1)$$

■

Lemma D.3. *Delta Expansion for $\hat{\lambda}_K$.*

$$\sqrt{K}(\hat{\lambda}_K - \lambda) = \frac{1}{\sqrt{K}} \sum_{k=1}^K L_k + o_p(1)$$

Proof of Lemma D.3. Define the map $g(w, b) = \frac{w}{b-w/N_p}$, then, $\lambda = g(\sigma, \theta_{N_p})$. Now,

$$\nabla g(\sigma, \theta_{N_p}) = \begin{pmatrix} g_w \\ g_b \end{pmatrix} \bigg|_{w=\sigma, b=\theta_{N_p}} = \begin{pmatrix} \frac{b}{(b-w/N_p)^2} \\ -\frac{w}{(b-w/N_p)^2} \end{pmatrix} \bigg|_{w=\sigma, b=\theta_{N_p}} = \begin{pmatrix} \theta_{N_p}/\rho^2 \\ -\sigma/\rho^2 \end{pmatrix}$$

By the Delta method,

$$\sqrt{K}(\hat{\lambda}_K - \lambda) = \nabla g(\sigma, \theta_{N_p})' \begin{pmatrix} \sqrt{K}(\hat{w}_K - \sigma) \\ \sqrt{K}(\hat{b}_K - \theta_{N_p}) \end{pmatrix} + o_p(1) = \frac{1}{\sqrt{K}} \sum_{k=1}^K L_k + o_p(1)$$

where the equality is exact for $\sqrt{K}(\hat{w}_K - \sigma)$ and $o_p(1)$ as per Lemma D.2 for $\sqrt{K}(\hat{b}_K - \theta_{N_p})$.

■

Corollary D.4. *Asymptotic Measurability of Design.* Let $\mathcal{G} = \sigma((L_k)_{k \geq 1})$ be the σ -algebra generated by the $(L_k)_{k \geq 1}$ sequence. Then, under Lemma C.1, the design (K^π, N^π) with $\pi \in \{\text{EtC-CS}, \text{ACS}\}$ is asymptotically $\mathcal{G}$ -measurable.

Lemma D.5. *Stable CLT.* Under Assumptions 6.1, for some fixed N ,

$$\frac{1}{\sqrt{K}} \sum_{k=1}^K M_{k,N} \xrightarrow[K \rightarrow \infty]{\mathcal{G}\text{-stably}} \sqrt{\theta_N} Z, \quad Z \sim N(0, 1), \quad Z \perp \mathcal{G},$$

Proof of Lemma D.5. The proof will rely on a (random index version of) Martingale Stable CLTs as in Hall and Heyde (2014). First, consider some fixed K . To invoke the relevant theorem we need

- **Martingale Difference Property:** Let $\mathcal{F}_0 = \mathcal{G}$ and $\mathcal{F}_k = \sigma((L_j)_{j \geq 1}, M_{1,N}, \dots, M_{k,N})$ such that $\mathcal{G} \subseteq \mathcal{F}_k$ for all k . Under cluster independence $\mathbb{E}[M_{k,N} \mid \mathcal{F}_{k-1}] = \mathbb{E}[M_{k,N} \mid L_k] = 0$ (by Lemma D.1). Thus $(M_{k,N}, \mathcal{F}_k)$ is a martingale difference sequence. By *frontloading* $\mathcal{G}$ in $\mathcal{F}$ we ensure that $\mathcal{F}$ -stability implies $\mathcal{G}$ -stability.
- **Conditional Variance:** $\frac{1}{K} \sum_k \mathbb{E}[M_{k,N}^2 \mid \mathcal{F}_{k-1}] = \frac{1}{K} \mathbb{E}[M_{k,N}^2 \mid L_k] \rightarrow \mathbb{E}[M_{k,N}^2] = \theta_N$ (by the LLN, provided $\theta_N < \infty$). Conditional on N , the variance is constant.
- **Lindeberg Condition:** Under Assumption 6.1.ii, (finite fourth moments) Lyapunov holds, i.e. $\frac{1}{K^2} \sum_k \mathbb{E}[M_{k,N}^4] \rightarrow 0$. Hence, Lindeberg condition holds.

Under the conditions above (see Hall and Heyde (2014) or Häusler and Luschgy (2015)), by the martingale stable CLT

$$\frac{1}{\sqrt{K}} \sum_{k=1}^K M_{k,N} \rightarrow N(0, \theta_N), \quad \mathcal{F}\text{-stably}$$

We can now replace fixed K by a random K^π (in an Anscombe style, see Häusler and Luschgy (2015)) by replacing K by κB , taking B asymptotics (instead of K) and noting that $K^\pi/B \xrightarrow{P} \kappa_0$ as per Lemma C.3. Finally, stable convergence with non-random limiting variance θ_N implies mixing convergence (see Aldous and Eagleson (1978)). And mixing convergence implies independence between limit and any $\mathcal{G}$ measurable process. ■

Proof of Theorem 6.2. Observe that conditional on K^π , N^π is fixed (via budget constraints). Moreover, $N^\pi \rightarrow N_0$ under Lemma C.3 so $\theta_N \rightarrow \theta_0$. Finally, under Corollary D.4 K^π is asymptotically $\mathcal{G}$ -measurable, therefore, under Lemma D.5

$$Z_B = \frac{\sqrt{K^\pi}(\hat{\mu}(K^\pi, N^\pi) - \mu)}{\sqrt{\theta_0}} \xrightarrow[B]{\mathcal{G}\text{-stably}} N(0, 1), \quad \text{with } N(0, 1) \perp K_A$$

The x -uniform part of the Theorem statement follows from Polya's Theorem (with $\Phi(x)$ continuous in x). The feasible $\hat{b}(K^\pi, N^\pi)$ statement follows by noting that $\hat{b}(K^\pi, N^\pi) \xrightarrow{P} \theta_0 > 0$ under Lemma C.3 and the CMT. Stable convergence is preserved under multiplication by a sequence converging in probability to a constant therefore,

$$Z_B \sqrt{\theta_0 / \hat{b}(K^\pi, N^\pi)} \xrightarrow{B} N(0, 1) \quad \mathcal{G}\text{-stably}$$

Coverage follows □

E Proofs of Section 7

E.1 Lower Bound Derivation

Policy Class. We first define the relevant policy class Π . The policy class is fairly general and contains most reasonable policies as especial cases. In particular, we consider policies which, in every period t , select a cluster $k_t = k$ and sample at least one unit $N_{k_t, t} \geq 1$ until they run out of budget. As a result, the game contains at most $T = K_m N_M$ rounds with strictly positive sampling. If no additional observation can be sampled without violating the budget in period t_M , $N_{k_t, t} = 0$ for all $t \geq t_M$. Importantly, the same cluster k can be revisited several times. Let $N_k(T) = \sum_{t=1}^T \mathbb{1}(k_t = k) N_{k_t, t}$ be the number of samples from cluster k up to period T . Define as well $K(T) = \sum_k \mathbb{1}(N_k(T) > 0)$. All policies must satisfy the budget constraint almost surely, i.e.

$$\mathbb{P}_\pi \left(\sum_{k \in [K(T)]} (F + V N_k(T)) \leq B \right) = 1$$

Data Generating Process. Consider two normally distributed Data Generating Processes (DGPs) y_-, y_+ indexed by ω_- and ω_+ , where $\omega_- = (\mu, \rho_-, \lambda_-)$ and $\omega_+ = (\mu, \rho_+, \lambda_+)$ where $\lambda_- = \lambda_0 - \delta$, $\lambda_+ = \lambda_0 + \delta$ for some $\lambda_0 \in (\lambda_m, \lambda_M)$ and a small positive δ . Additionally, $\rho_- = \sigma / \lambda_-$ and $\rho_+ = \sigma / \lambda_+$. Under the additive errors model in Equation 1

$$y_{+ki} = \mu + u_{+k} + \varepsilon_{ki}$$

where $u_{+k} \sim N(0, \rho_+)$. y_{-ki} and u_{-k} are defined as expected.

Separating Events. As it is common practice in lower bound derivations, we characterize a separating event. Following the characterization of the variance for the generalized game (where $N_k \neq N$ for all k), we write the separating event in terms of deviations from the harmonic mean N_H . To do so, we need to characterize (or at least lower bound) the excess variance in terms of the harmonic mean. By the linearity of the cost structure $K_T(F + \sum_{k \in [K_T]} V N_k(T)) = K_T(F + V \bar{N}(T))$ where $\bar{N} = \frac{1}{K_T} \sum_{k \in [K_T]} N_k(T)$ is the arithmetic mean. We know that $N_H(T) \leq \bar{N}(T)$, hence $\frac{1}{K_T} \geq \frac{F + V N_H}{B}$. It follows

$$W(\mathbf{N}(T), \omega) = \frac{\rho}{K_T(\mathbf{N}(T))} \left(1 + \frac{\lambda}{N_H(\mathbf{N}(T))} \right) \geq \frac{\rho(F + VN_H(T))}{B} \left(1 + \frac{\lambda}{N_H(T)} \right) = W_H(N_H(T), \omega)$$

Now, take the environment $\omega = \omega_+$, let $N_+ = N^*(\lambda_+)$ and consider the event $N_H \leq N_0 = N^*(\lambda_0)$. Under the remarks above, the excess variance of a policy which selects $\mathbf{N}(T)$ such that $N_H \leq N_0$ can be lower bounded by

$$W(\mathbf{N}(T), \omega_+) - W(N_+, \omega_+) \geq W_H(N_H, \omega_+) - W_H(N_+, \omega_+) \geq W_H(N_0, \omega_+) - W_H(N_+, \omega_+)$$

where we used the fact that $W(N^*, \omega) = W_H(N^*, \omega)$ because the oracle allocation is a balanced one (under a simple convexity argument). A symmetric result holds for ω_- . Finally, by noting that the expression for W_H is completely equivalent to the expression for $W(K^*(N), N, \omega)$ in Appendix A, one can recover a second order Taylor approximation of the form

$$W_H(N_0, \omega_+) - W_H(N_+, \omega_+) = \frac{\rho_0 \sqrt{FV}}{4B\lambda_0^{3/2}} (\lambda_0 - \lambda_+)^2 + O((\lambda_0 - \lambda_+)^3/B)$$

Augmented Experiment and Divergence Decomposition. For some policy $\pi \in \Pi$, let $(\tilde{\mathbb{P}}_{\pi-}, \tilde{\mathbb{P}}_{\pi+})$ be the probability measures induced by the interaction between the policy π and the environments y_- and y_+ , respectively. Now, consider the augmented experiment where, every time the learner samples cluster k , she not only observes $(y_{ki})_{N_k}$ but also u_k . Let $(\mathbb{P}_{\pi-}, \mathbb{P}_{\pi+})$ be the probability measures induced by the interaction between the policy π and the augmented environments. By data processing,

$$D_{\text{KL}}(\tilde{\mathbb{P}}_{\pi-}, \tilde{\mathbb{P}}_{\pi+}) \leq D_{\text{KL}}(\mathbb{P}_{\pi-}, \mathbb{P}_{\pi+})$$

The augmented game simplifies the analysis of the KL-divergence significantly, as revisiting a cluster does not add any more *cluster-level* KL. This fact, together with the zero KL contribution of individual errors (provided they have the same distribution in both environments), allows us to decompose $D_{\text{KL}}(\mathbb{P}_{\pi-}, \mathbb{P}_{\pi+})$ in the usual *bandit style*,²⁵ i.e.

$$D_{\text{KL}}(\mathbb{P}_{\pi-}, \mathbb{P}_{\pi+}) = \mathbb{E}_{\pi-}[K_T] D_{\text{KL}}(N(0, \rho_-), N(0, \rho_+))$$

We further observe that for all policies $\pi \in \Pi$, $FK_T + \sum_k VN_k(T) \leq B$ a.s., thus $K_T \leq B/F$ a.s.. The expression above incorporates the fact that all clusters within each environment are equally distributed. Next

²⁵See Chapter 15 in [Lattimore and Szepesvári \(2020\)](#) for a complete treatment.

$$\begin{aligned}
D_{\text{KL}}(N(0, \rho_-), N(0, \rho_+)) &= \frac{1}{2} \left[\frac{\rho_-}{\rho_+} - 1 - \ln \left(\frac{\rho_-}{\rho_+} \right) \right] = \frac{1}{2} \left[\frac{\lambda_+}{\lambda_-} - 1 - \ln \left(\frac{\lambda_+}{\lambda_-} \right) \right] \\
&= \frac{1}{2} \phi \left(\frac{\lambda_+}{\lambda_-} \right) = \frac{1}{2} \phi \left(1 + \frac{2\delta}{\lambda_0 - \delta} \right)
\end{aligned}$$

where the third equality serves as definition of $\phi(x) = x - 1 - \ln(x)$. For $\delta \leq \lambda_0/2$, and using the fact that $\phi(1+x) \leq x^2/2$

$$\frac{1}{2} \phi \left(\frac{\lambda_+}{\lambda_-} \right) \leq \frac{1}{2} \phi \left(1 + \frac{4\delta}{\lambda_0} \right) \leq \frac{4\delta^2}{\lambda_0^2}$$

Therefore,

$$D_{\text{KL}}(\tilde{\mathbb{P}}_{\pi-}, \tilde{\mathbb{P}}_{\pi+}) \leq \frac{B}{F} \frac{4\delta^2}{\lambda_0^2} = C_1 B \delta^2$$

The argument is now completed through a standard Bretagnolle-Huber argument

$$\begin{aligned}
&\tilde{\mathbb{E}}_{\pi-} [W(K, \mathbf{N}, \omega_-)] - W(K^*(\lambda_-), N^*(\lambda_-), \omega_-) + \tilde{\mathbb{E}}_{\pi+} [W(K, \mathbf{N}, \omega_+)] - W(K^*(\lambda_+), N^*(\lambda_+), \omega_+) \geq \\
&\tilde{\mathbb{E}}_{\pi-} [W_H(N_H, \omega_-)] - W_H(N_-, \omega_-) + \tilde{\mathbb{E}}_{\pi+} [W_H(N_H, \omega_+)] - W_H(N_+, \omega_+) \geq \\
&\tilde{\mathbb{P}}_{\pi-} \left(N_H > \sqrt{\lambda_0 \frac{F}{V}} \right) \cdot \left(\mathbb{E}_{\pi-} \left[W_H(N_H, \omega_-) \mid N_H > \sqrt{\lambda_0 \frac{F}{V}} \right] - W_H(N_-, \omega_-) \right) + \\
&\quad \tilde{\mathbb{P}}_{\pi+} \left(N_H \leq \sqrt{\lambda_0 \frac{F}{V}} \right) \cdot \left(\mathbb{E}_{\pi+} \left[W_H(N_H, \omega_+) \mid N_H \leq \sqrt{\lambda_0 \frac{F}{V}} \right] - W_H(N_+, \omega_+) \right) = \\
&\frac{\rho_0 \sqrt{FV}}{4B\lambda_0^{3/2}} \delta^2 \left(\tilde{\mathbb{P}}_{\pi-} \left(N_H > \sqrt{\lambda_0 \frac{F}{V}} \right) + \tilde{\mathbb{P}}_{\pi+} \left(N_H \leq \sqrt{\lambda_0 \frac{F}{V}} \right) \right) + O(\delta^3/B) \geq \\
&\frac{\rho_0 \sqrt{FV}}{4B\lambda_0^{3/2}} \delta^2 \frac{1}{2} \exp \left(-D_{\text{KL}}(\tilde{\mathbb{P}}_{\pi-}, \tilde{\mathbb{P}}_{\pi+}) \right) - \frac{C_3 \delta^3}{B} \geq \\
&\frac{\rho_0 \sqrt{FV}}{4B\lambda_0^{3/2}} \delta^2 \frac{1}{2} \exp \left(-D_{\text{KL}}(\mathbb{P}_{\pi-}, \mathbb{P}_{\pi+}) \right) - \frac{C_3 \delta^3}{B} \geq \\
&\frac{\rho_0 \sqrt{FV}}{8B\lambda_0^{3/2}} \delta^2 \exp \left(-C_1 \delta^2 B \right) - \frac{C_2 \delta^3}{B}
\end{aligned}$$

Where the first inequality follows from our variance relaxation under the harmonic depth, the second inequality is a simple event partition, the third inequality follows from the second order Taylor expansion anticipated in the Section, the fourth inequality follows from Bretagnolle-Huber inequality, the fifth inequality from a data processing argument, and the final inequality from our derivations above. Finally, by selecting $\delta = \sqrt{1/(C_1 B)}$, which is smaller than $\lambda_0 \leq \lambda/2$ for sufficiently large B , we recover

$$\frac{\rho_0 \sqrt{FV}}{8B\lambda_0^{3/2}} \frac{1}{C_1 B} \exp(-1) - \frac{C_2 \delta^3}{B} \geq \frac{C_3}{B^2} - \frac{C_4}{B^{5/2}}$$

for some finite and strictly positive C_3 . Result follows from applying the inequality $\max(a, b) \geq \frac{a+b}{2}$ and absorbing it (as well as the higher order terms) into a constant C . $\square$

E.2 Policy Comparison

Proof of Lemma 7.2. The first part of the Lemma is immediate, and follows from a simple second order approximation around λ_0 . See Lemma C.2 for a complete derivation. For the minmax optimal policy, recall that

$$\begin{aligned} W(K^*(\lambda), N^*(\lambda), \omega_0) - W(K^*(\lambda_0), N^*(\lambda_0), \omega_0) &= \frac{\rho_0}{B} \left[F + V\lambda_0 + \sqrt{FV} \left(\sqrt{\lambda} + \frac{\lambda_0}{\sqrt{\lambda}} \right) \right] - \frac{\rho_0 (\sqrt{F} + \sqrt{V\lambda_0})^2}{B} \\ &\quad - \frac{\rho_0 \sqrt{FV}}{B} \cdot \frac{(\sqrt{\lambda} - \sqrt{\lambda_0})^2}{\sqrt{\lambda}} \end{aligned}$$

Hence, the maximum is achieved at $\rho_0 = \rho_M$ and $\lambda_0 \in \{\lambda_m, \lambda_M\}$ (whichever yields a higher loss). Finally, the minimum is obtained by

$$\min_{\lambda} \max \left\{ \frac{(\sqrt{\lambda} - \sqrt{\lambda_M})^2}{\sqrt{\lambda}}, \frac{(\sqrt{\lambda} - \sqrt{\lambda_m})^2}{\sqrt{\lambda}} \right\}$$

which yields $\lambda_{mM} = \left(\frac{\sqrt{\lambda_m} + \sqrt{\lambda_M}}{2} \right)^2$. Finally, the same second order approximation returns

$$W(K^*(\lambda_{mM}), N^*(\lambda_{mM}), \omega_0) - W(K^*(\lambda_0), N^*(\lambda_0), \omega_0) = \frac{C(\lambda_{mM} - \lambda_0)^2}{B} + O\left(\frac{|\lambda_{mM} - \lambda_0|^3}{B}\right) \sim O(1/B)$$